Migrating Application to AWS

AWS Data Migration Services

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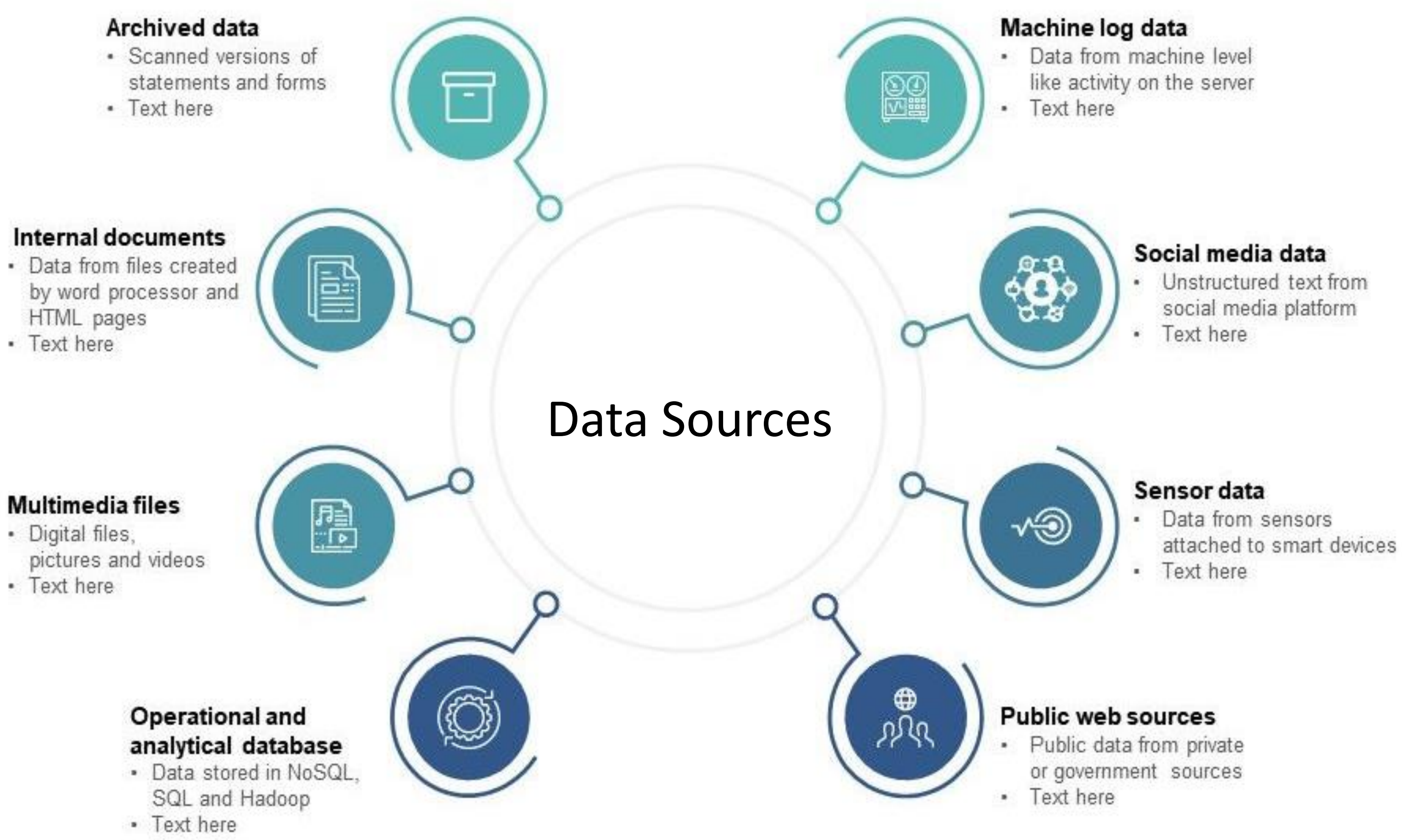

1000s G,T,P,E,Z,Y

By 2020, there will be around 40 trillion gigabytes of data (40 zettabytes)

AWS Data Migration Services

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Challenges

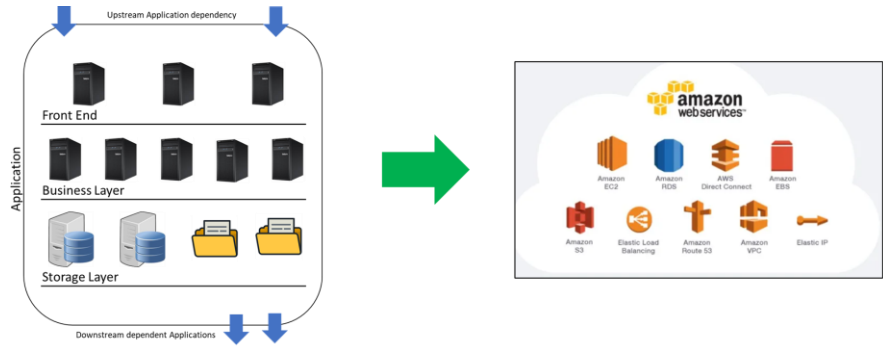
Moving terabytes or even petabytes of data to the cloud is a big challenge. This is a time-consuming task to move all that data to cloud. While companies plan and migrate this large scale of data to cloud, following factors must be considered seriously.

Data Storage
Pushing data into cloud without considering how that is going to be used. Do not consider cost alone as a primary factor, consider its usage too. Object storage(S3) vs File storage (EBS, EFS/FSx) needs careful consideration basis how they will be used by application. Consider changing File storage be Object storage after a period of time?
This may be different for each use case.

Data Preparation
Filter out unwanted data. Prioritize what data should move first basis application that needs it. Same data may be required in different format when in cloud. Compression and archiving of data, aging definition or retention period for each dataset that is moving to cloud.

Validation
Integrity checking is the single most important step, and the easiest to get wrong. Use pessimistic approach, assume corruption will occur during data transfer. Employ checksums before and after. Do these checks before migrating applications/systems are turned off for good.

Transfer Marshalling
If data preparation involves copying, exporting, reformatting, or archiving, local storage can become a bottleneck. It may be necessary to set up dedicated local storage to stage the prepared data. This allows many systems to perform data preparation in parallel before the media is shipped.



The common cloud data migration challenge

Type	Speed in mbps
T1	1.544
T2	6.312
T3	44.736

AWS Data Migration Services

Direct Connect

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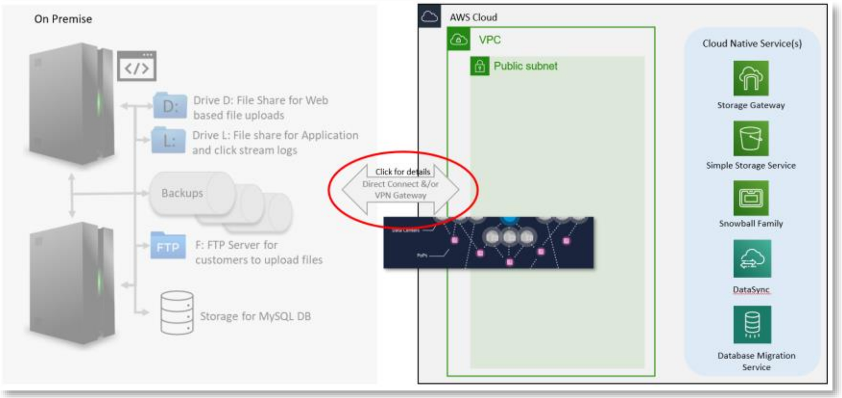

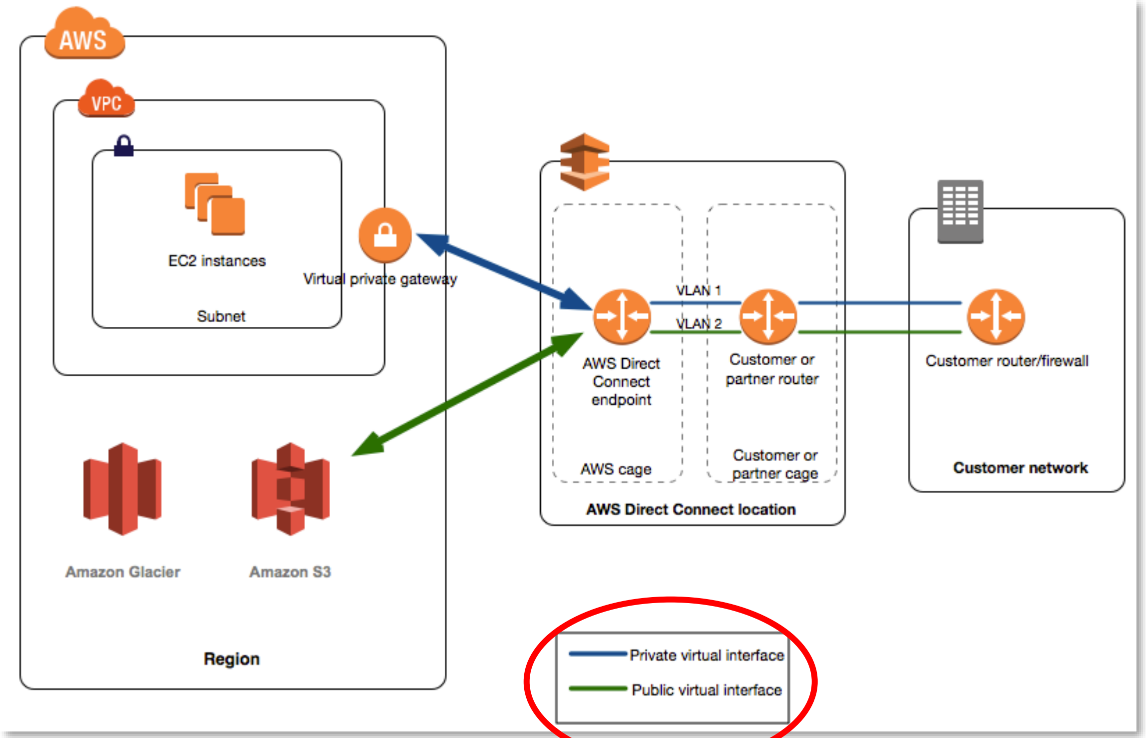

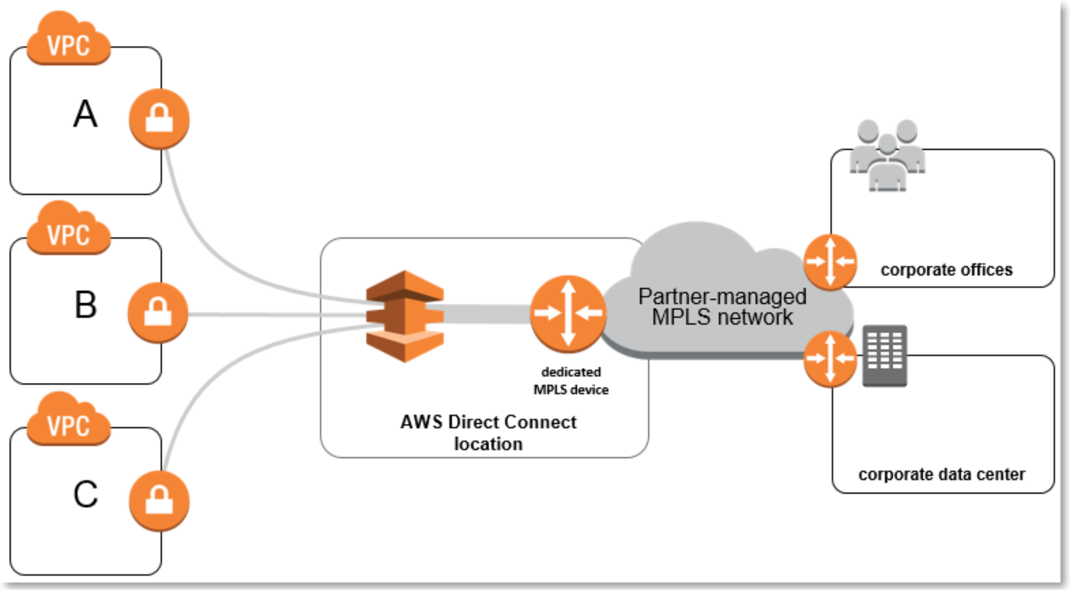

Benefits
REDUCES YOUR BANDWIDTH COSTS
CONSISTENT NETWORK PERFORMANCE
COMPATIBLE WITH ALL AWS SERVICES
PRIVATE CONNECTIVITY TO YOUR AMAZON VPC
ELASTIC

AWS Data Migration Services   Direct Connect

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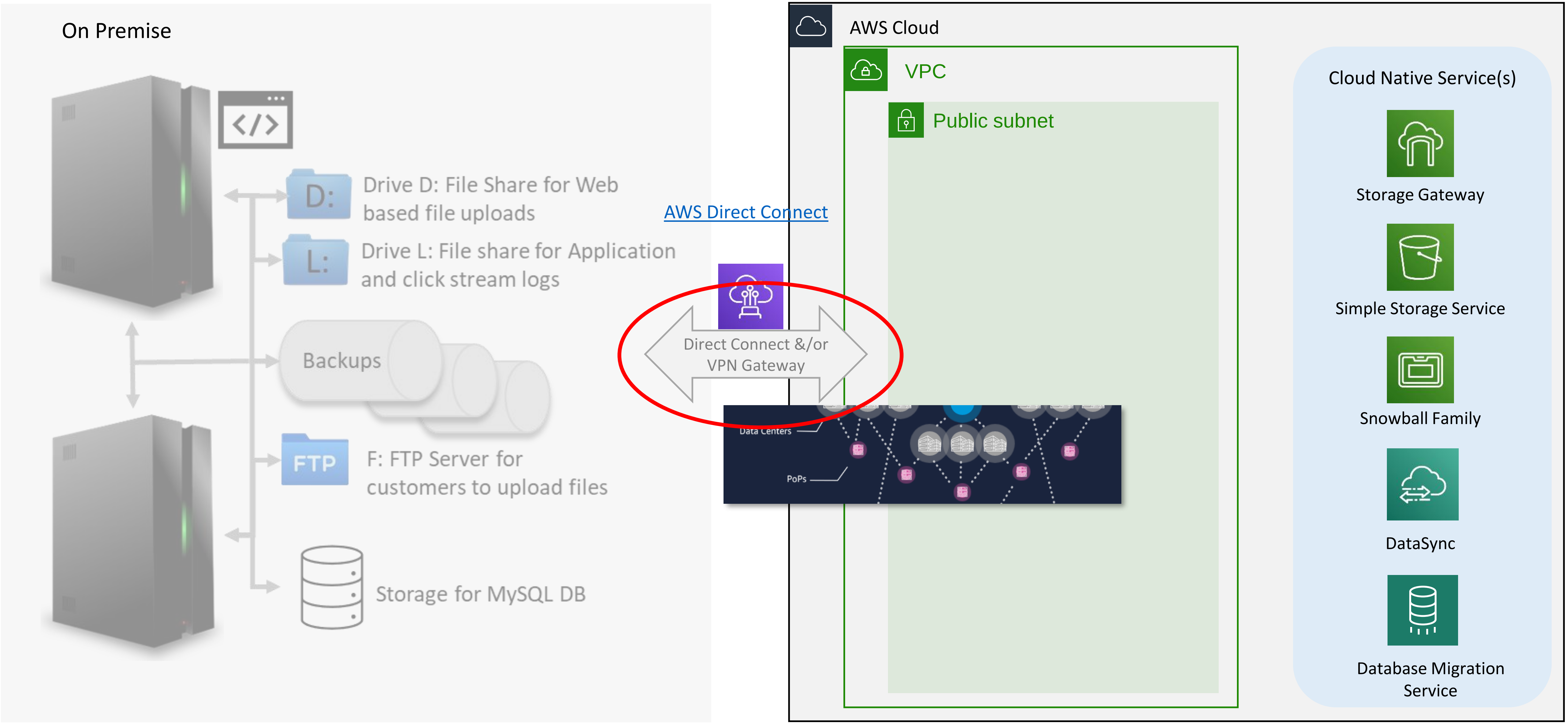
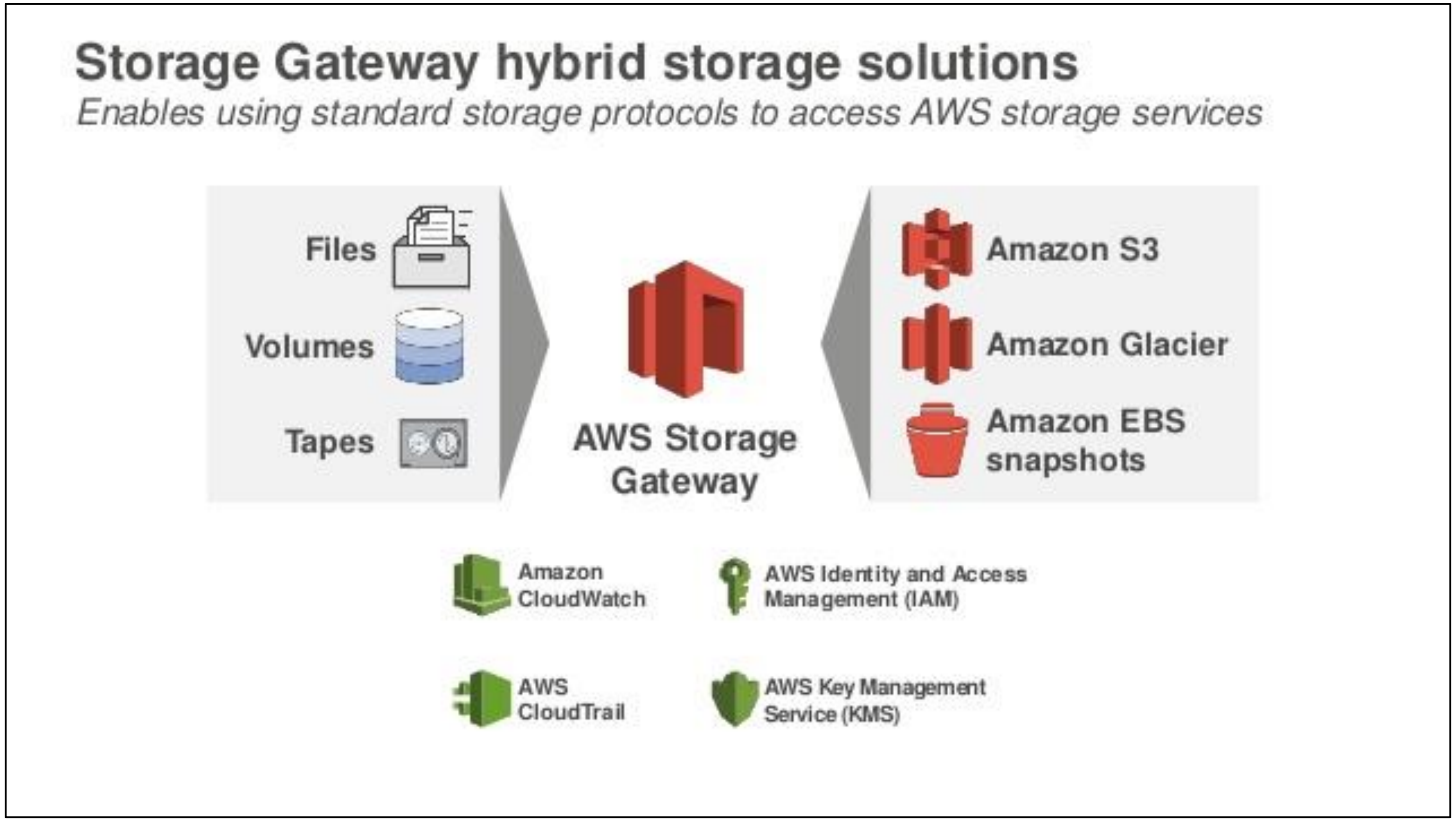


Good alternative for Bastion Host → How to setup Pritunl VPN on AWS to Access Servers

AWS Data Migration Services [Storage Gateway]

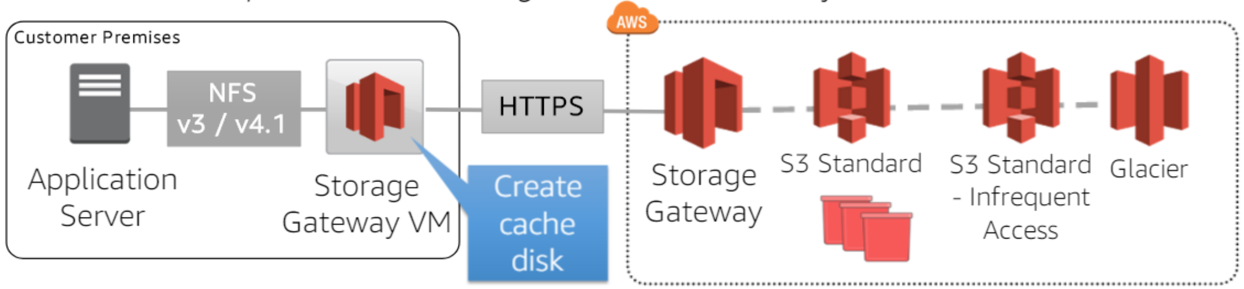


Benefits
Integrated, Performance, Optimized transfers, Cloud Scale, Durable and secure
Hybrid Cloud workloads - Big data analytics, data processing, machine learning or cloud data migration workloads with on-premises applications or data sources require architectures that require both local access and a connection to a central cloud object storage repository - Amazon S3
Disaster recovery on AWS, Backup and Restore etc.

[Good overview of Storage Gateway Service]

AWS Data Migration Services [Storage Gateway]

On-premises file storage maintained as objects in Amazon S3



- Data stored and retrieved from your Amazon S3 buckets
- Use lifecycle policies, versioning, or cross region replication to manage data
- One-to-one mapping from files-to-objects
- File metadata stored in object metadata
- Bucket access managed by IAM role you own and manage

- File system metadata persisted in object user-metadata, eg.

```
x-amz-meta-file-permissions: 0666
x-amz-meta-file-user-class: 4321
x-amz-meta-file-group-class: 42
x-amz-meta-file-created: 2016-10-05T20:08:45+00:00
x-amz-meta-file-last-modified: 2016-10-05T20:08:45+00:00
```

[Storage Gateway Limits]

[Storage Gateway Hardware Appliance]

AWS Data Migration Services [Storage Gateway]

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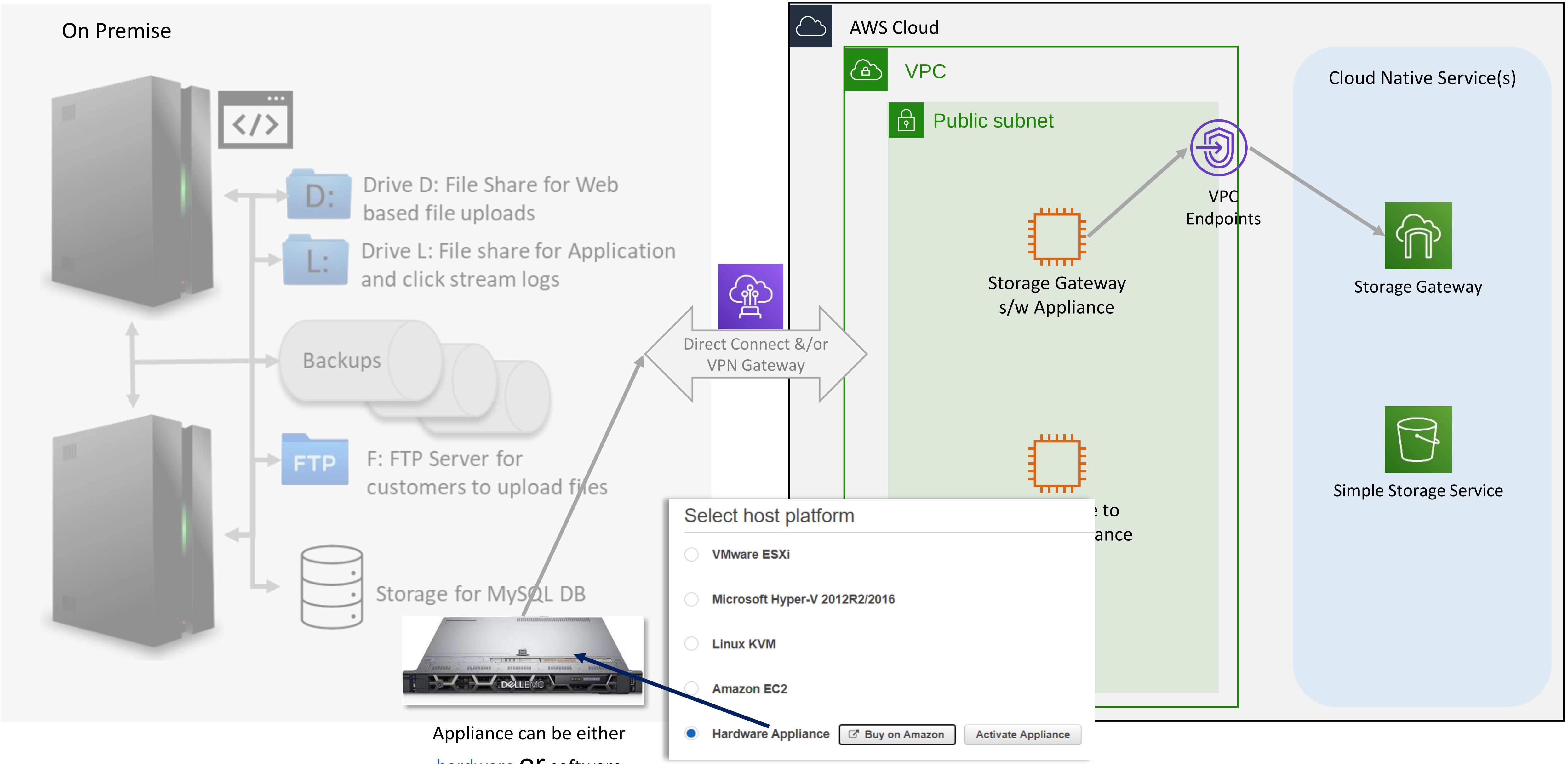

Appliance can be either [hardware] or software

On-Prem VMWare Virtualization

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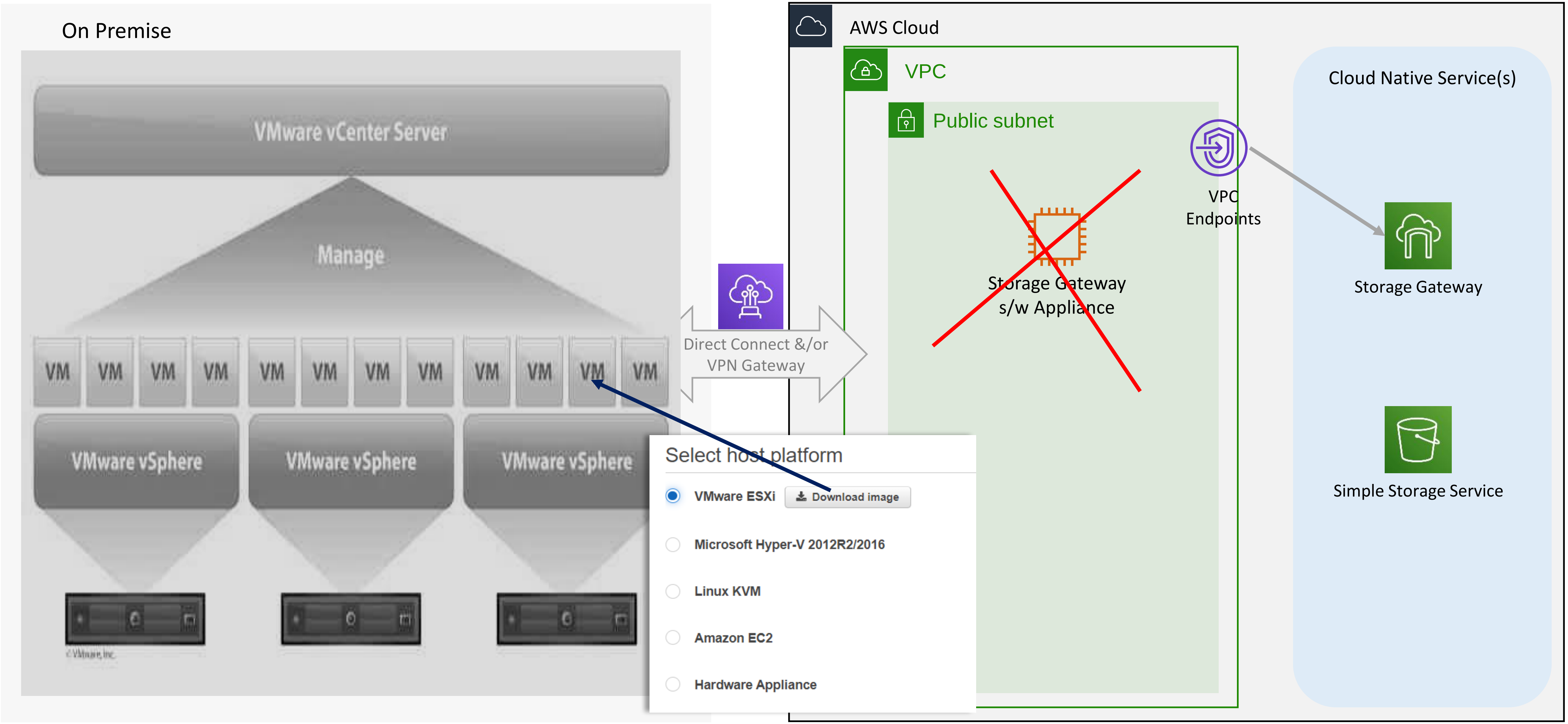

On-Prem Hyper-V Virtualization

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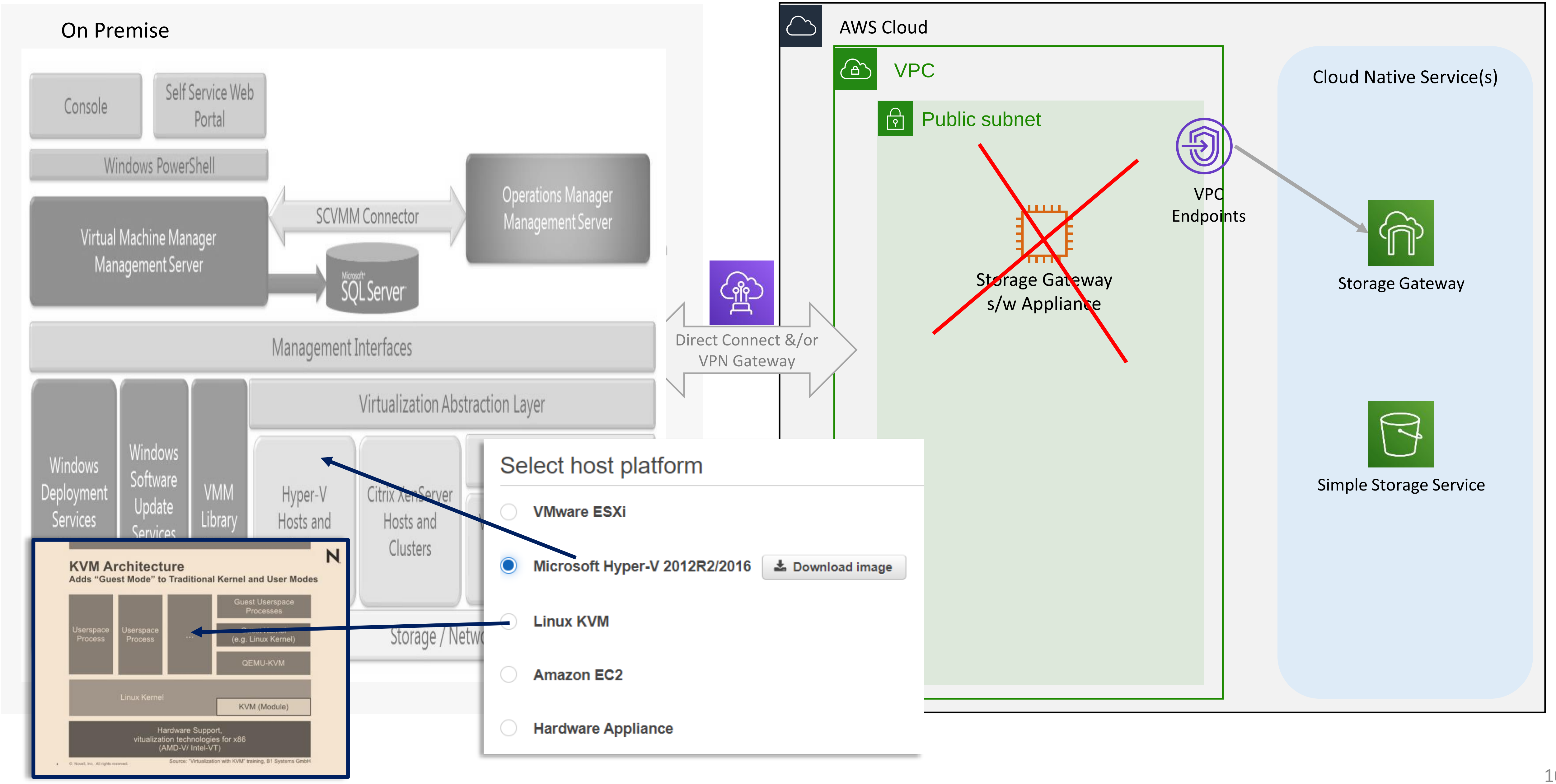

AWS Data Migration Services [Storage Gateway]

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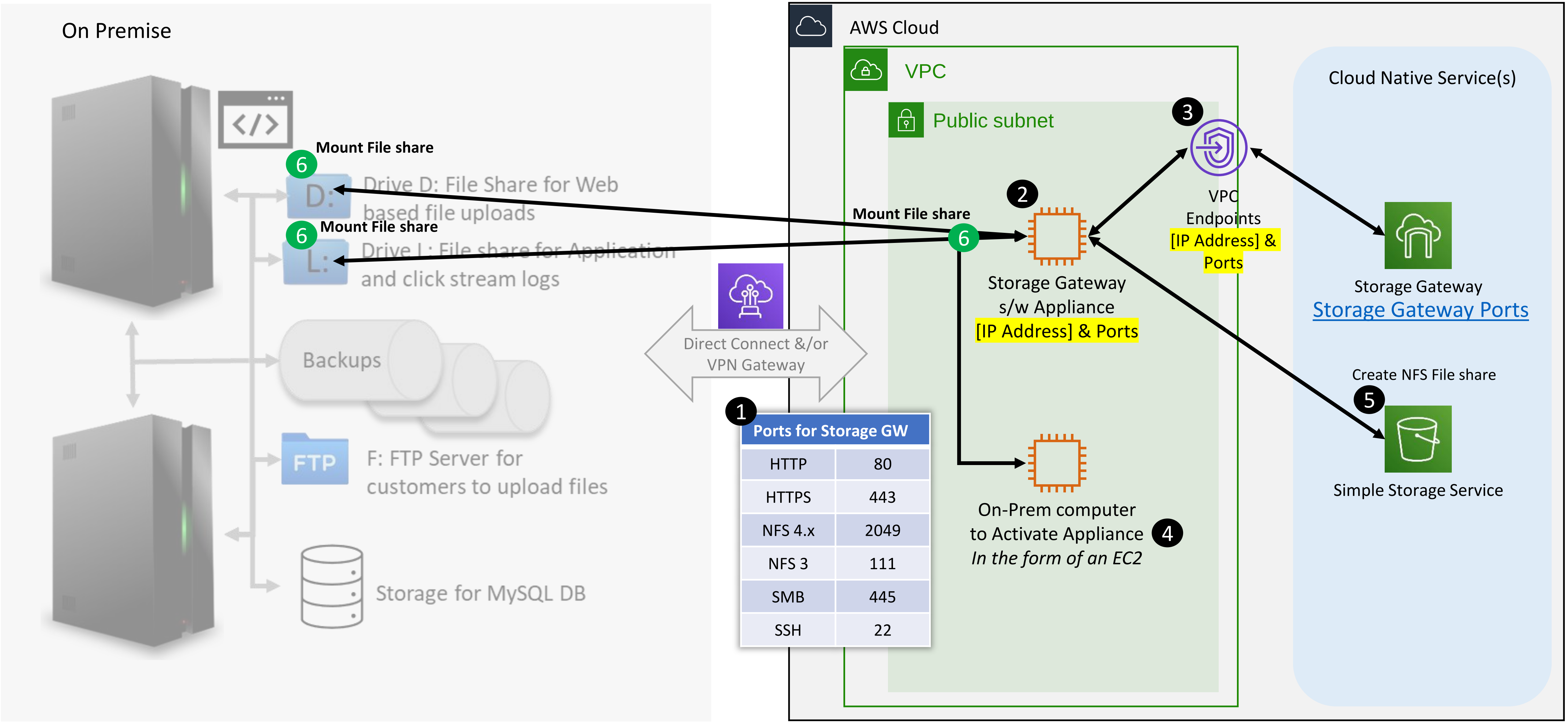
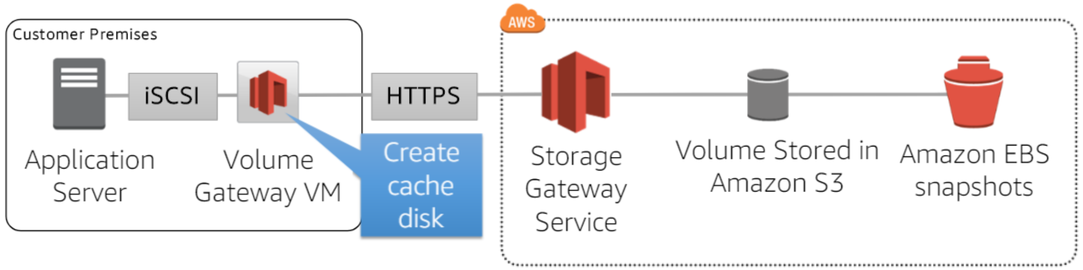


Ports for Storage GW	
HTTP	80
HTTPS	443
NFS 4.x	2049
NFS 3	111
SMB	445
SSH	22

AWS Data Migration Services

Storage Gateway

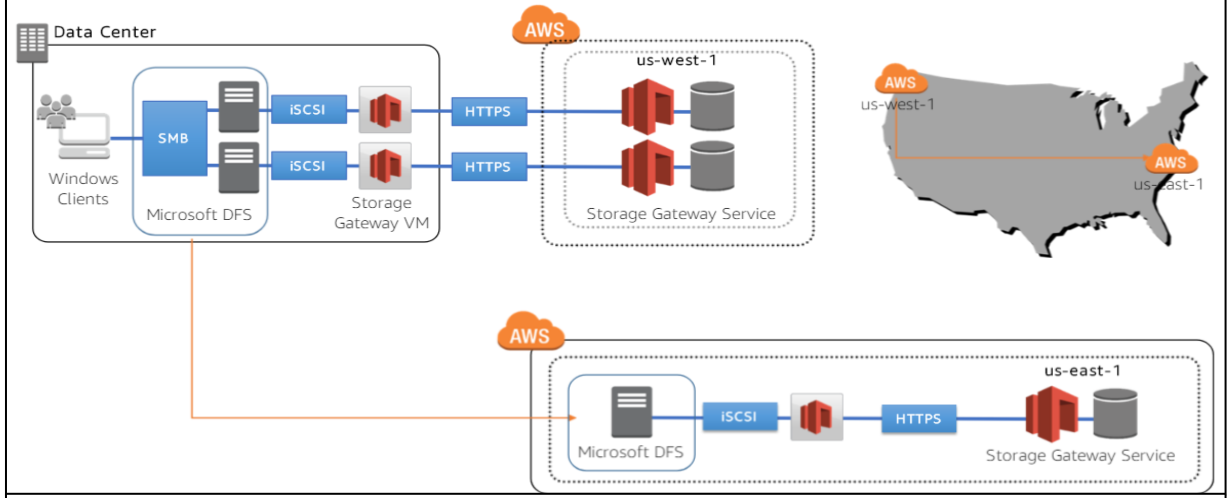
On-premises volume storage backed by Amazon S3 with EBS snapshots

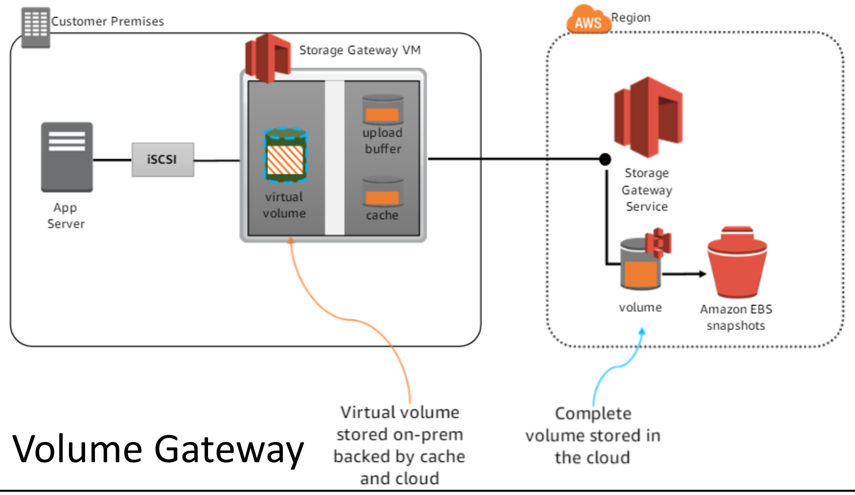


- Up to 1PB of total volume storage per gateway
- Block storage in Amazon S3 accessed via the volume gateway
- Compression of data in-transit and at-rest
- Backup on-premises volumes to EBS snapshots
- Create on-premises volumes from EBS snapshots

Volume Gateway

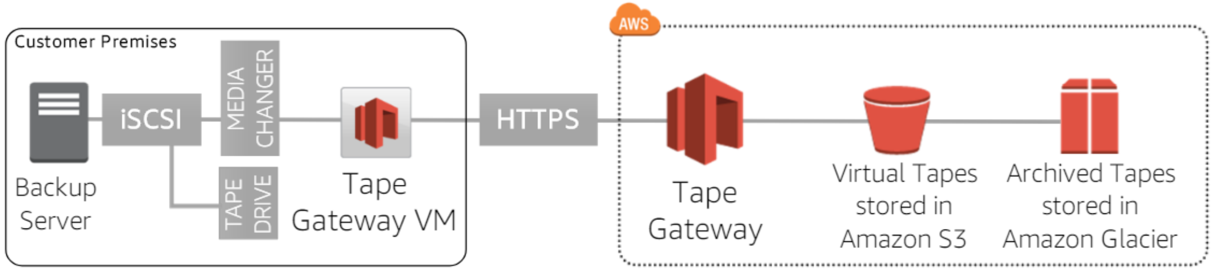
Local HA with remote disaster recovery





Cached Volume Gateway

Virtual tape storage in Amazon S3 and Amazon Glacier with VTL management



- Virtual tape storage in S3 and Glacier accessed via tape gateway
- Up to 1 PB total tape storage per gateway, unlimited archive capacity
- Supports leading backup applications
- Data compressed in-transit and at-rest

Tape Gateway

AWS Data Migration Services DataSync

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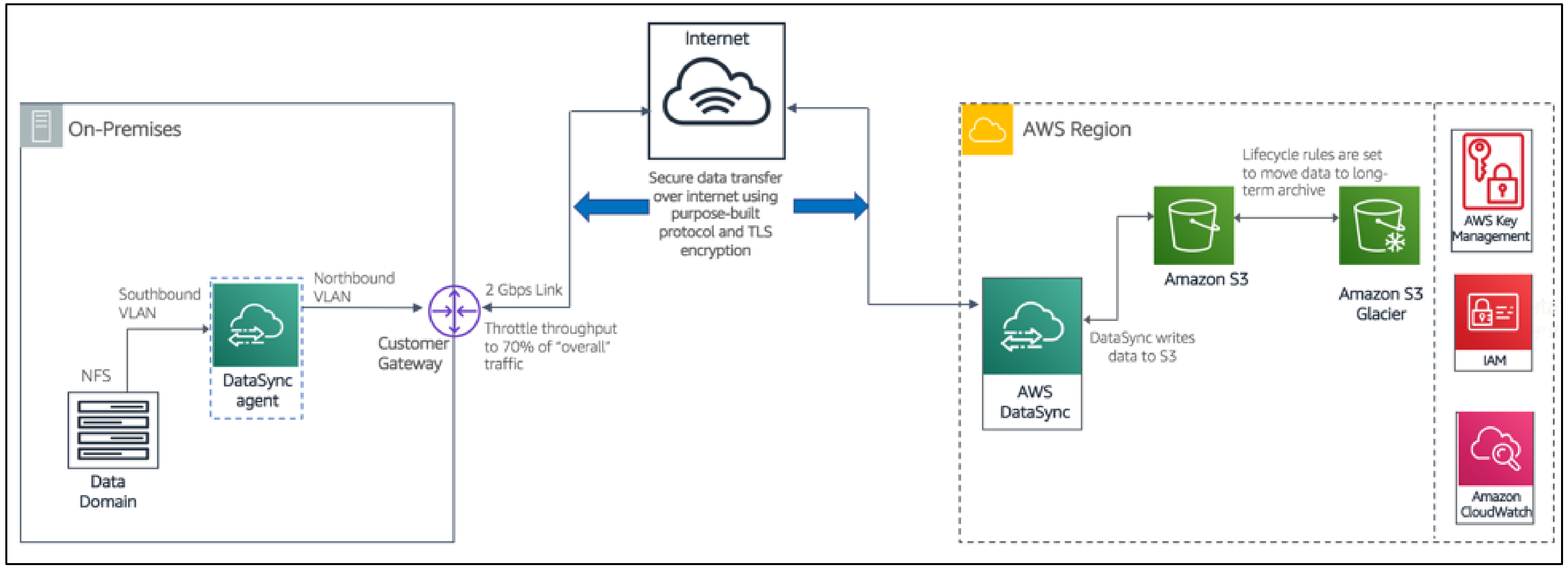

Benefits
Automate data transfer from on-premise to storage to S3 and EFS at 10x speed
Handles most of the heavy lifting related to running instance, handling encryption, managing scripts, network optimization and data integrity validation
Uses on-premise software agent to connect to existing storage (NFS/SMB) protocols
Transfer one-time data load, recurring data processing workflow & replication

AWS Data Migration Services DataSync

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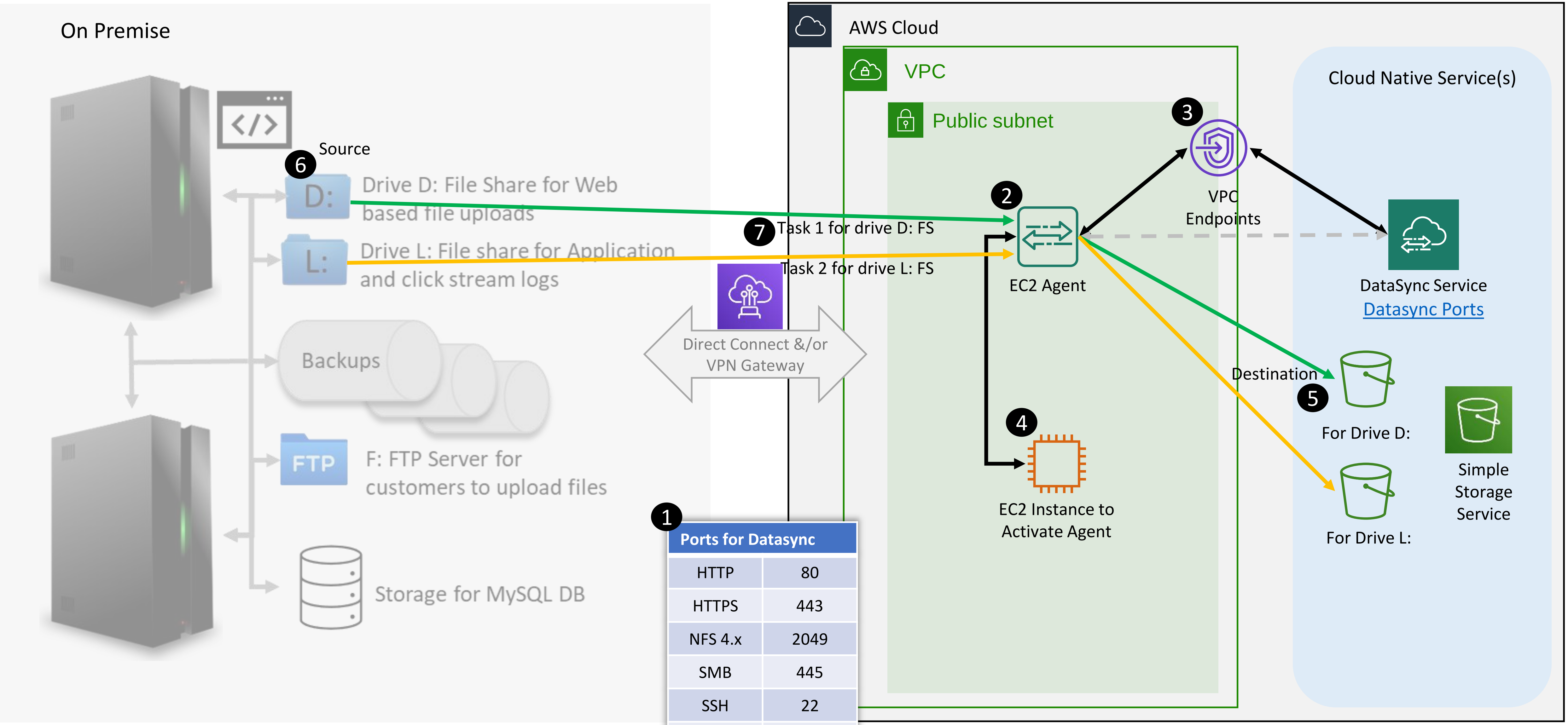

Ports for Datasync	
HTTP	80
HTTPS	443
NFS 4.x	2049
SMB	445
SSH	22
VPC endpoint	1024-1064

aws-datasync-1567007105 (ami-015f1e7b12f3d0326) – Not sure, lost my agent after activating

AWS Data Migration Services

S3 Transfer Acceleration

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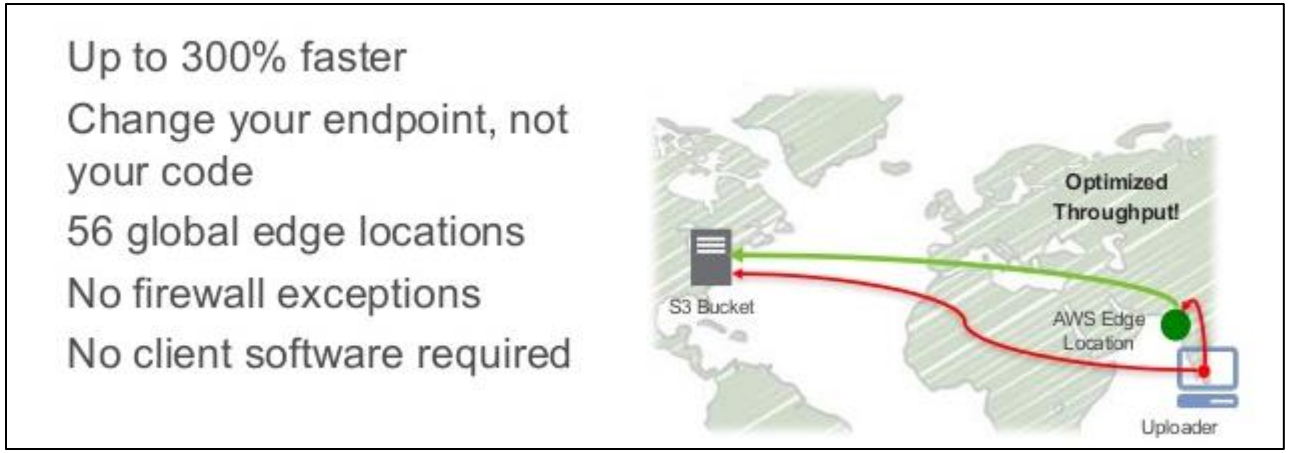
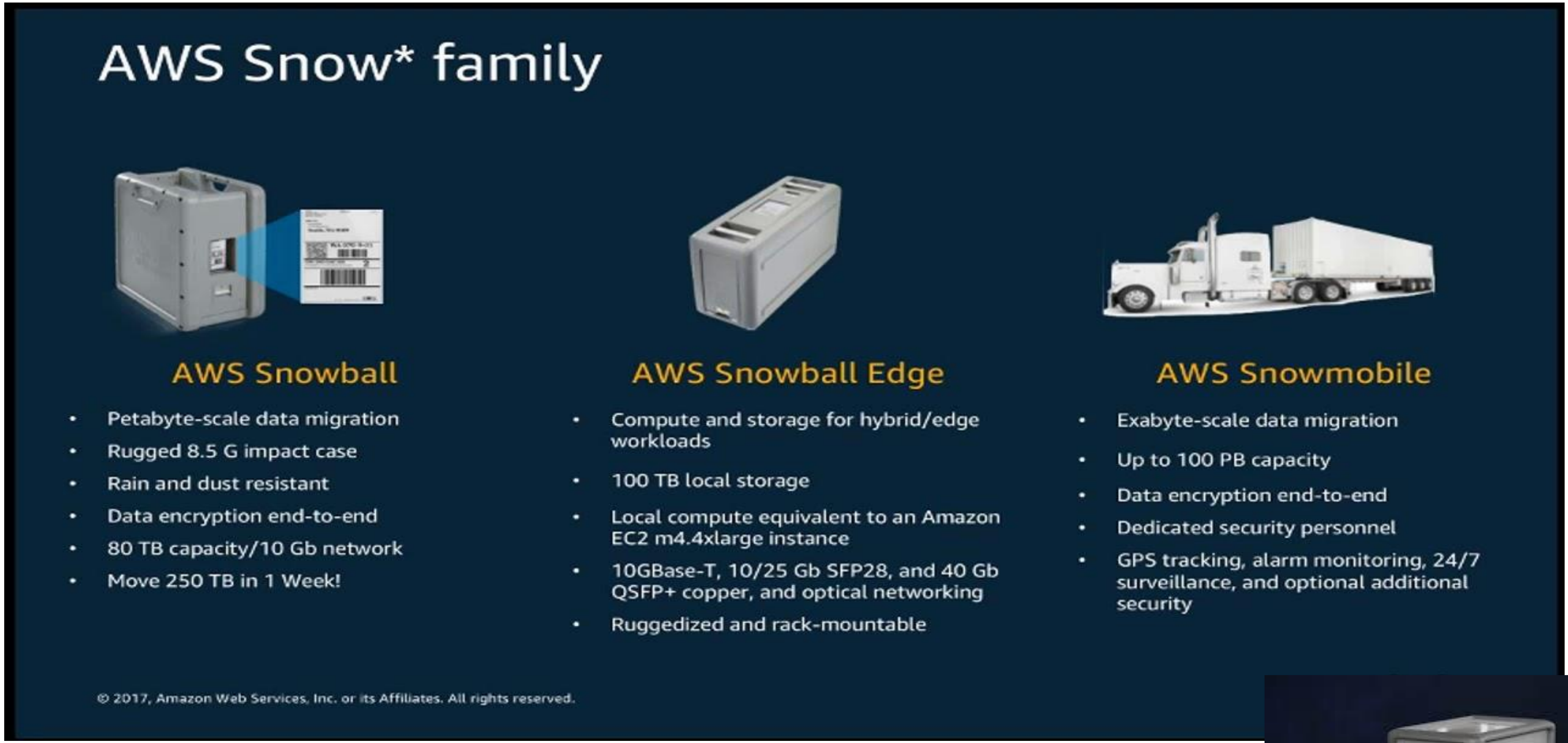


Click here to see live demo

Benefits
Accelerates Amazon S3 data transfers by making use of optimized network protocols and the AWS edge infrastructure. Improvements are typically in the range of 50% to 500% for cross-country transfer of larger objects, but can go even higher under certain conditions
Make use of AWS edge network which has points of presence in more than 50 locations
Transferring data across or between continents, have a fast Internet connection, use large objects, or have a lot of content to upload.

Other Options - AWS Data pipeline & AWS Glue

AWS Data Migration Services [On-site Solutions - Snow family]

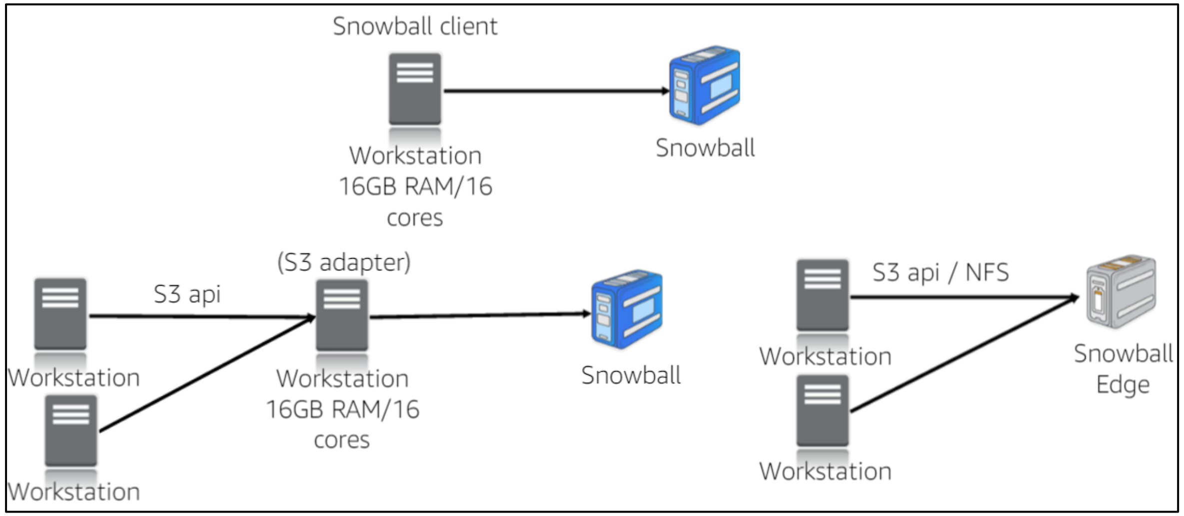


Benefits
Snowball is a petabyte-scale data transport solution that uses secure appliances to transfer large amounts of data into and out of the AWS cloud
Helps eliminate challenges that can be encountered with large-scale data transfers including high network costs, long transfer times, and security concerns
Check availability in the target Region
Use for import data to cloud or export data from cloud at computing locations

[Snowball Edge Process Flow]

[Snowcone → 8TB capacity 4.5 lbs., 9"L x 6"W x 3"H]
blog posted on 17-Jun-2020

On-site Solutions - Snow family



AWS Transfer Family SFTP, FTPS, FTP

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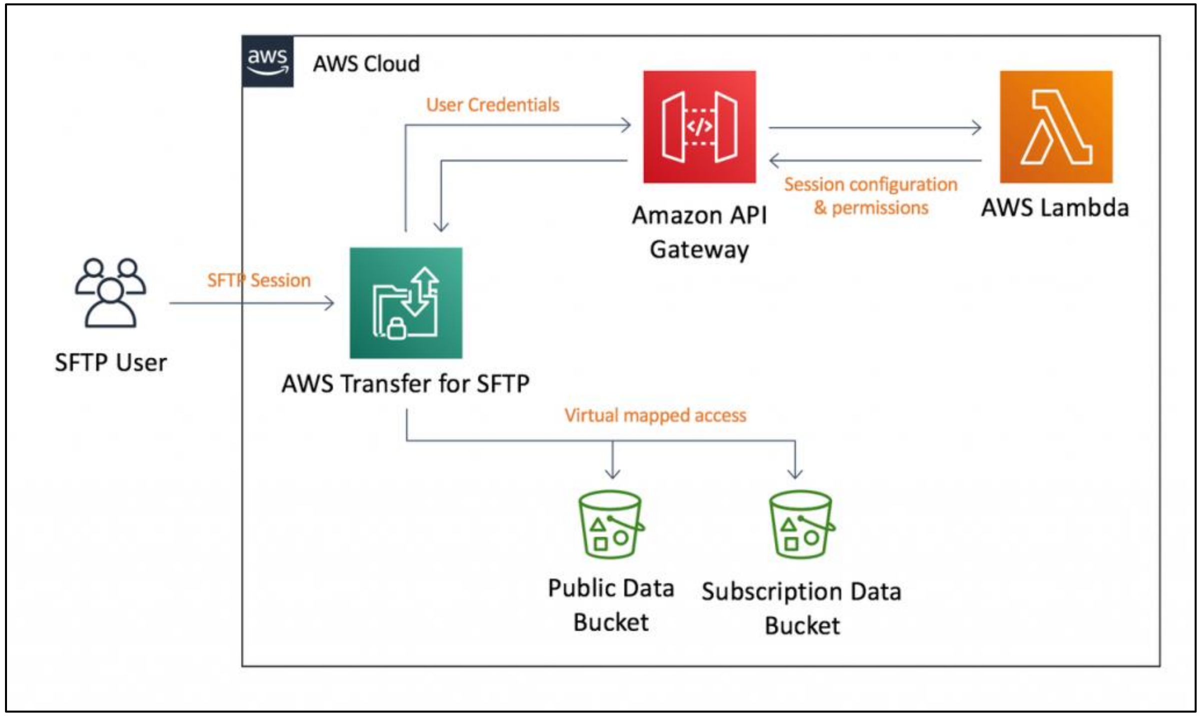

Benefits
Reduce operational burdens
Streamline workflow migrations
Integrate processing and management
API can be enabled on service for programmatic file transfers
If S3 is used to host static web site, this transfer service can help you manage contents without exposing AWS console to the third party or resource involved

AWS Transfer Family

[SFTP, FTPS, FTP]

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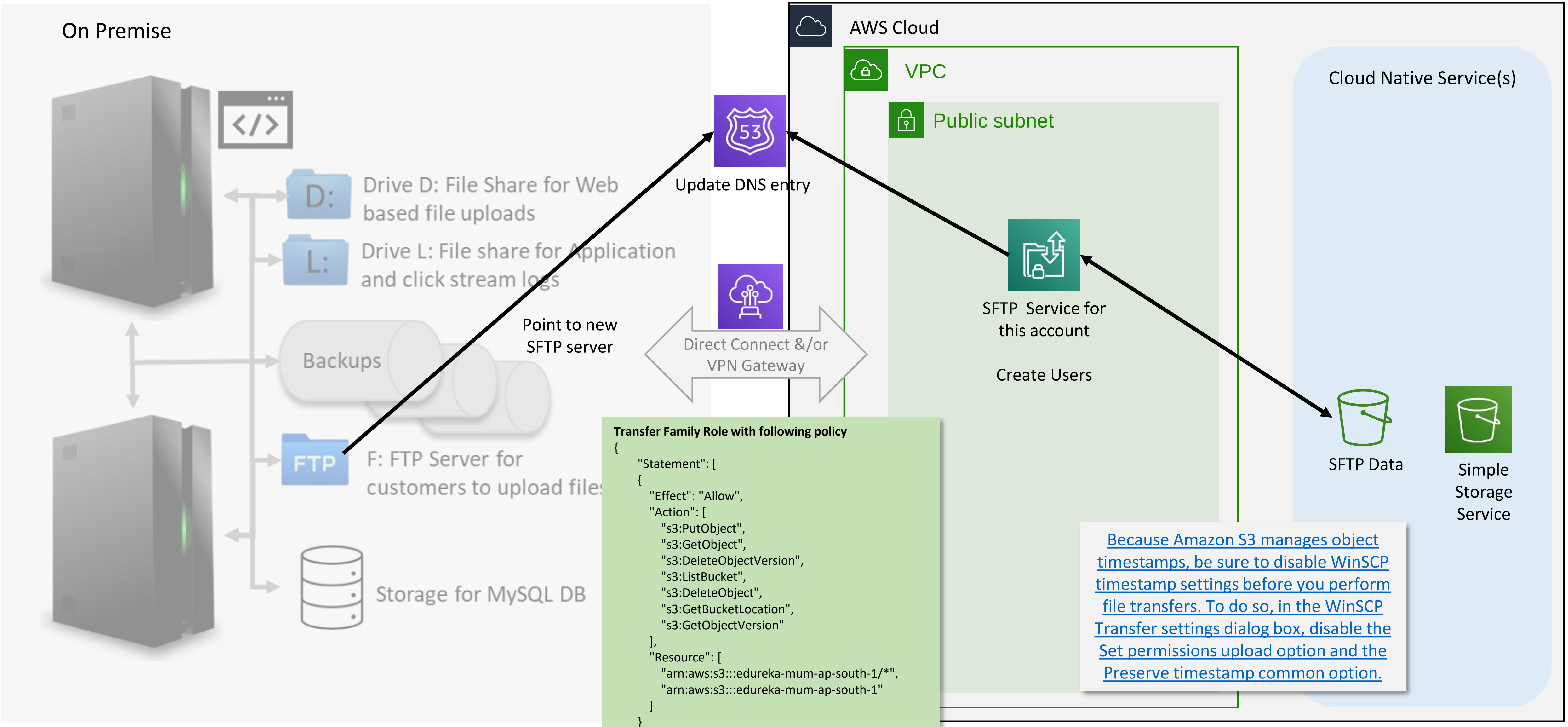


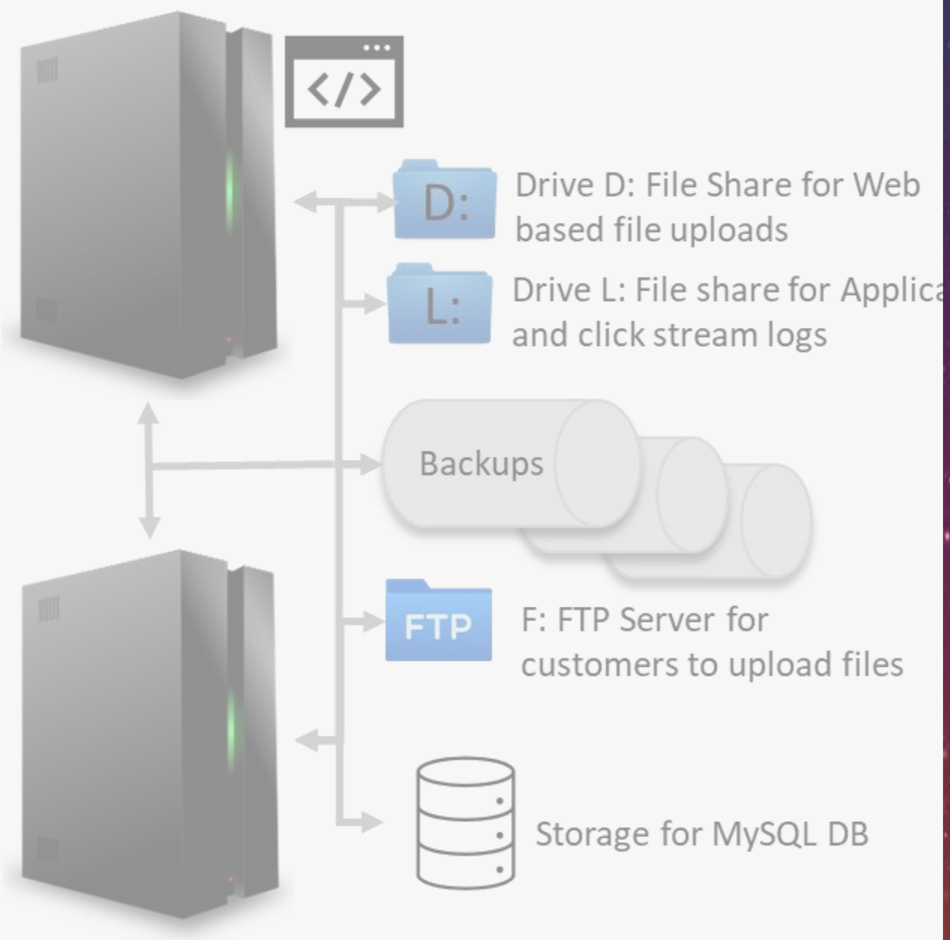
Transfer Family Role with following policy

```
{
    "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject",
        "s3:DeleteObjectVersion",
        "s3:ListBucket",
        "s3:DeleteObject",
        "s3:GetBucketLocation",
        "s3:GetObjectVersion"
      ],
      "Resource": [
        "arn:aws:s3:::edureka-mum-ap-south-1/*",
        "arn:aws:s3:::edureka-mum-ap-south-1"
      ]
    }
  ]
}
```

[Because Amazon S3 manages object timestamps, be sure to disable WinSCP timestamp settings before you perform file transfers. To do so, in the WinSCP Transfer settings dialog box, disable the Set permissions upload option and the Preserve timestamp common option.]

AWS Outposts

On Premise



AWS Remote Work and Learning

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- Tools for employees to work securely from any location
- AWS remote work and learning solutions allow employees, contact center agents, and students to remain productive and connected while working from home

Benefits
Provide Secure Cloud Desktops for Remote Employees → AWS WorkSpaces With Amazon WorkSpaces, your remote workers get a fast, responsive desktop of their choice that they can access anywhere, anytime, from any supported device
Enable Remote Learning and Working on legacy client server applications → AWS AppStream fully managed application streaming service that allows students to access the applications they need on any computer, whether they're in the classroom, the library, a cafe, or at home
Secure Content Collaboration → AWS WorkDocs makes it easy to collaborate with others from anywhere and lets you easily share content, provide rich feedback, and collaboratively edit documents
Secure and Scalable Remote Network Access → AWS Client VPN fully-managed, pay-as-you-go VPN service that elastically scales to support thousands of remote users. Client VPN is designed so your employees can access any company resource, both within AWS and on premises, from any location, with one console to manage all connections

ELB – Elastic Load Balancing

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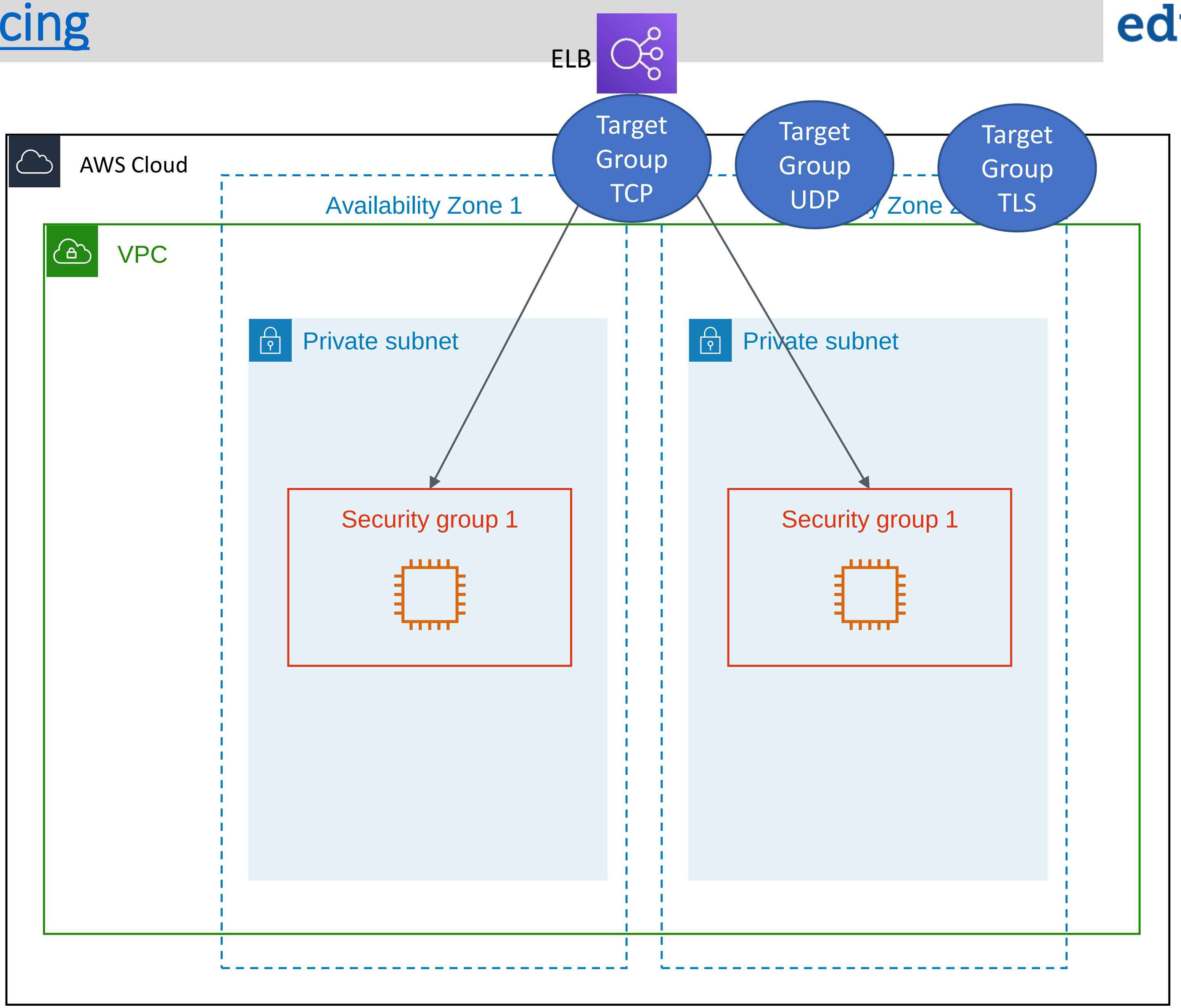
- ELB Types
  - Application ELB
  - Network ELB
- Listener
- Target Group(s)



Application Load Balancer – Routing options

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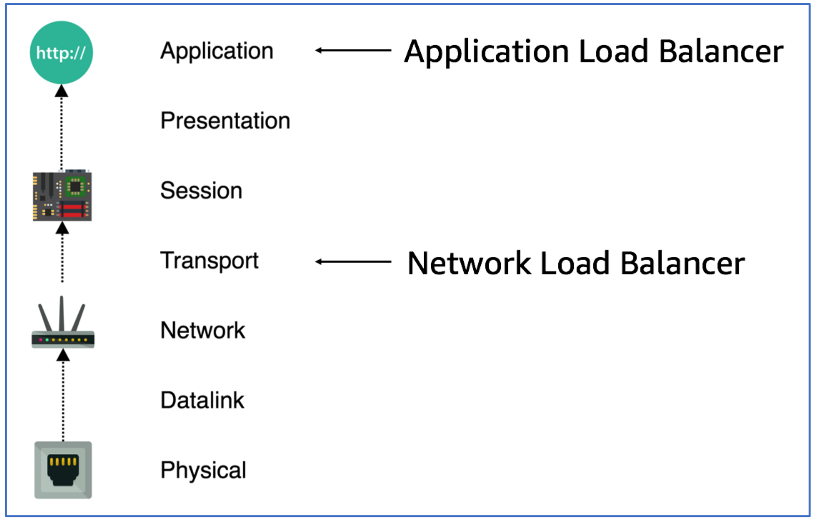
#	Routing Option	Route Client Request based on
1	Host-based	HOST field in HTTP header
2	Path-based	URL path on HTTP header
3	HTTP header-based	Value of any standard field or custom HTTP header
4	HTTP method-based	Any standard or custom HTTP method
5	Query String parameter-based	Query String or Query Parameter
6	Source IP address CIDR-based	Source IP address CIDR from where the request originates

Difference between ALB and NLB with example



Auto Scaling Group (ASG)

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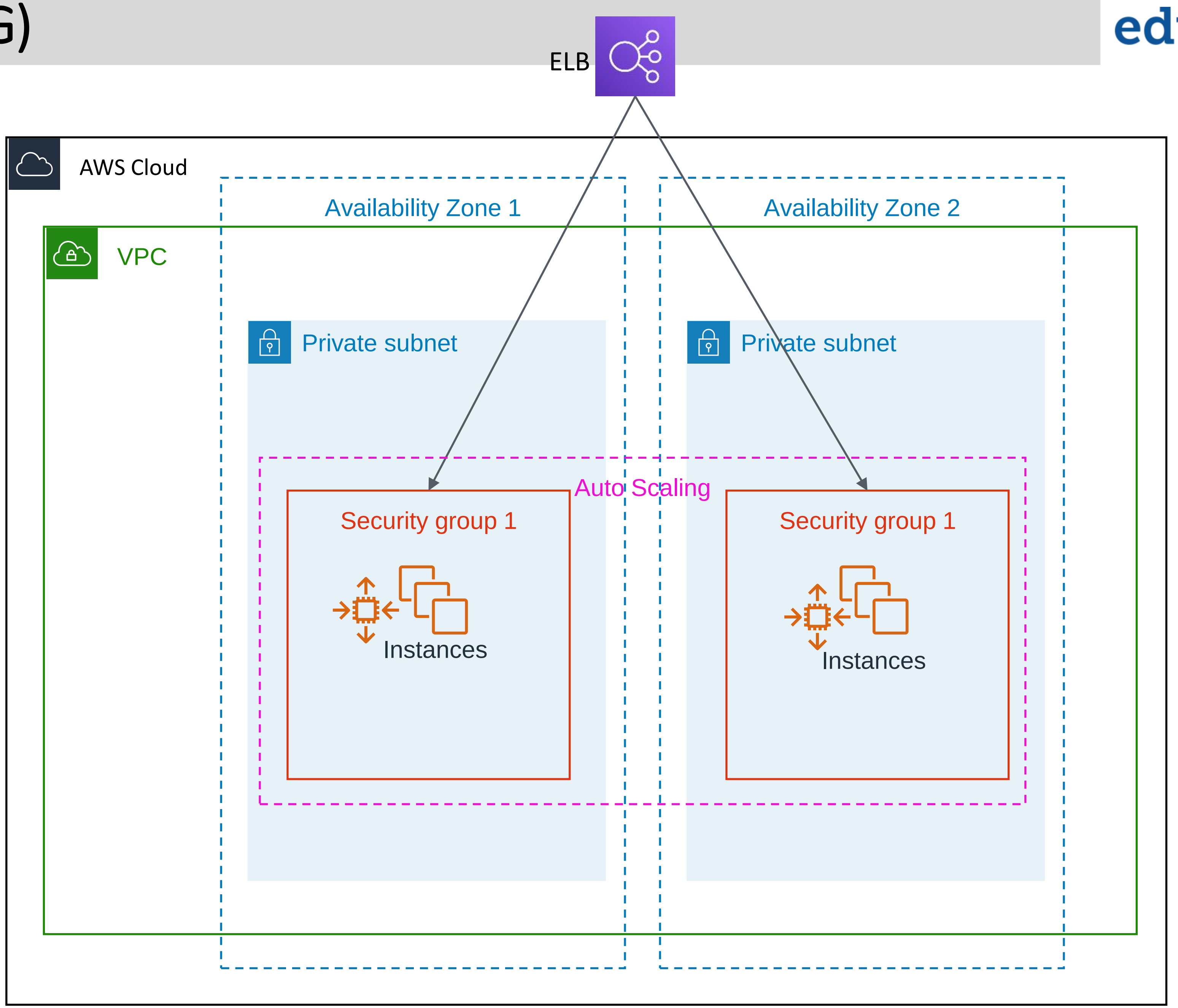
- ELB →TG → ASG
    - Horizontal Scaling
    - Scaling Rules
- Vertical Scaling
- How ASG lowers cost

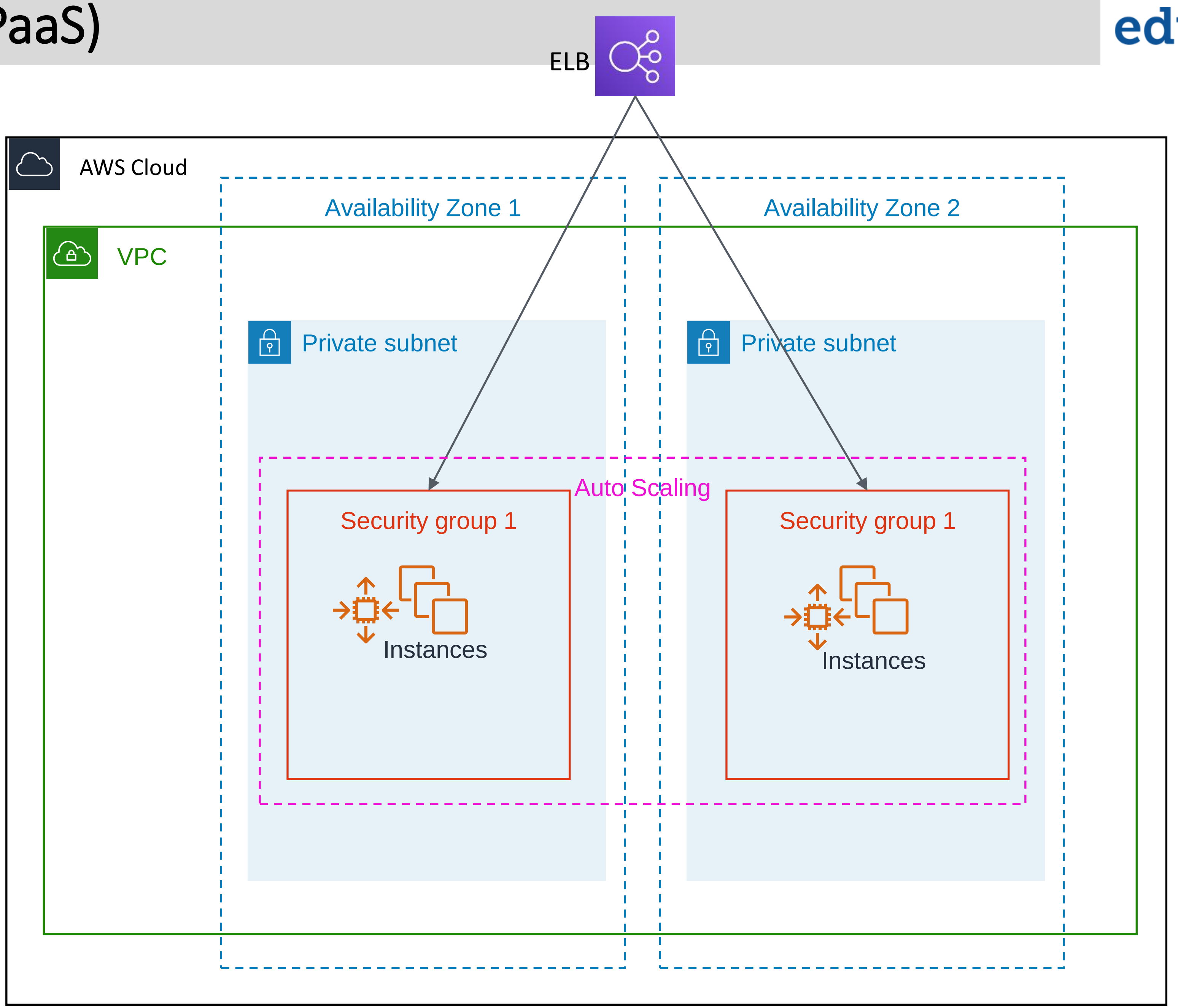


AWS Elastic Beanstalk (PaaS)

- Auto Scaling
- Load Balancing
- Monitoring
- Logging
- Tracing
- Deployment Options
- Application Platforms
- Customization
- Web, Batch, Containers

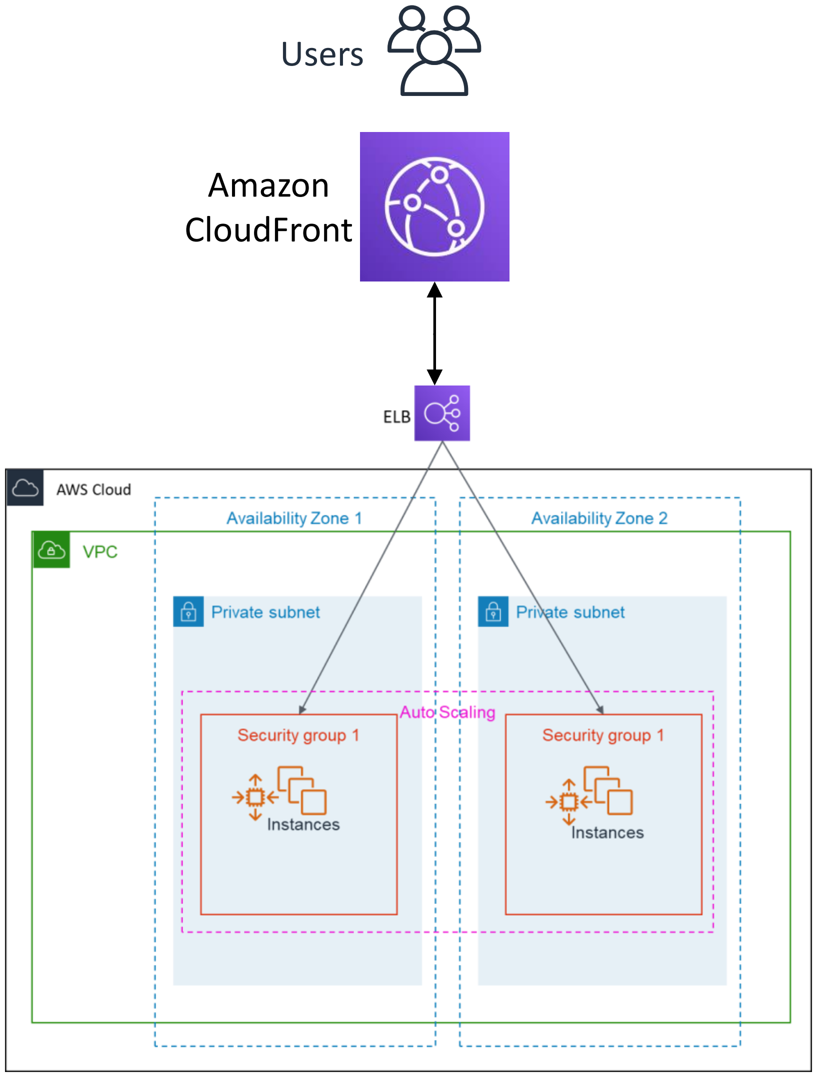
CLI Support

- Phyton >= 3.0
- pip install awsebcli



AWS CloudFront

- CDN – Content Delivery Network
- Extend ELB endpoint to CloudFront for boosting performance



EC2 pricing

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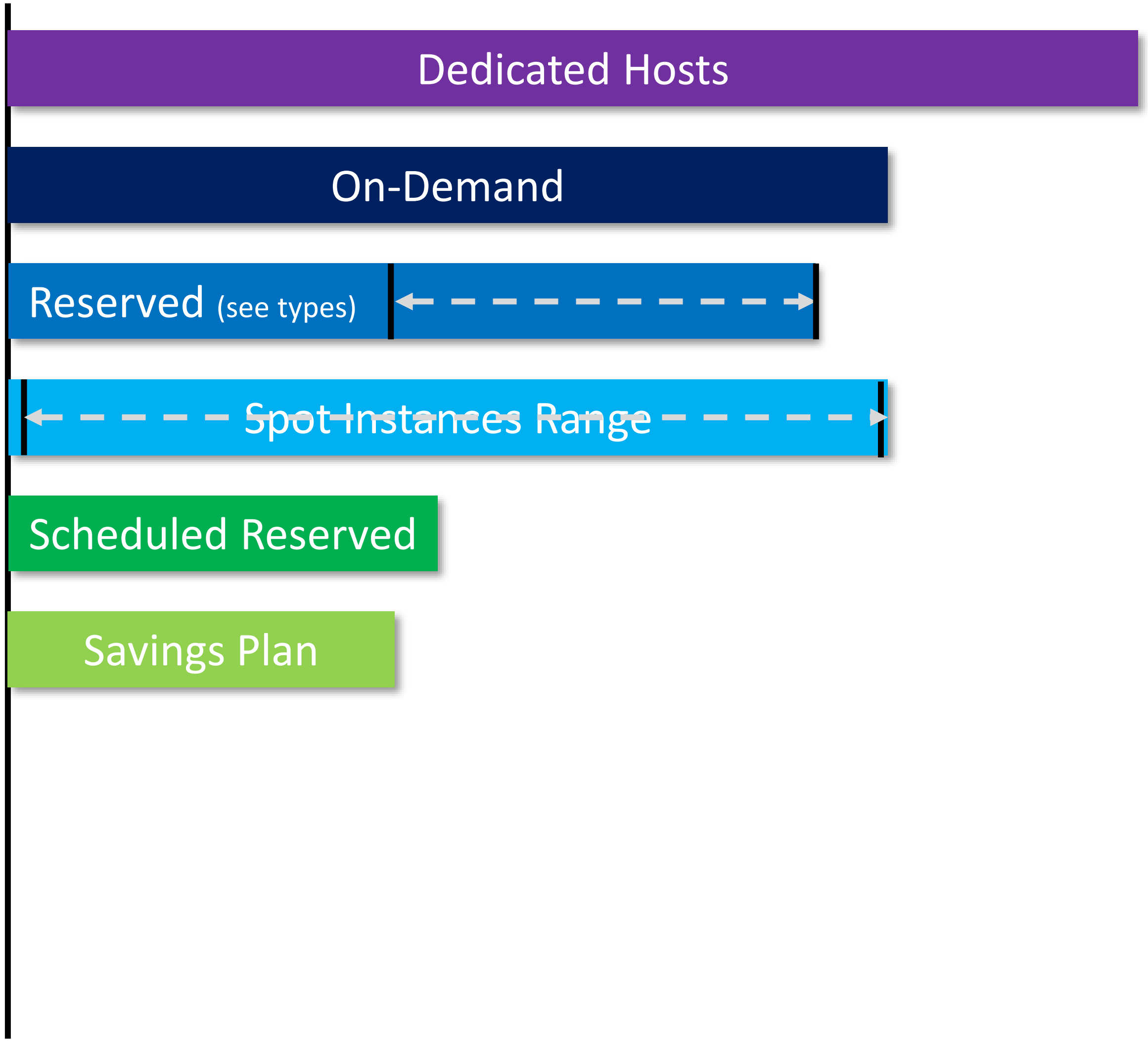
- Dedicated Hosts
- On-Demand
- Reserved Instances
- Spot Instances
- Scheduled Reserved Instances
  - Not available in all regions yet
- Savings Plan

AWS Cost Estimation Tool

Cost comparison using one EC2 Instance Type



Application Migration Process and Tools

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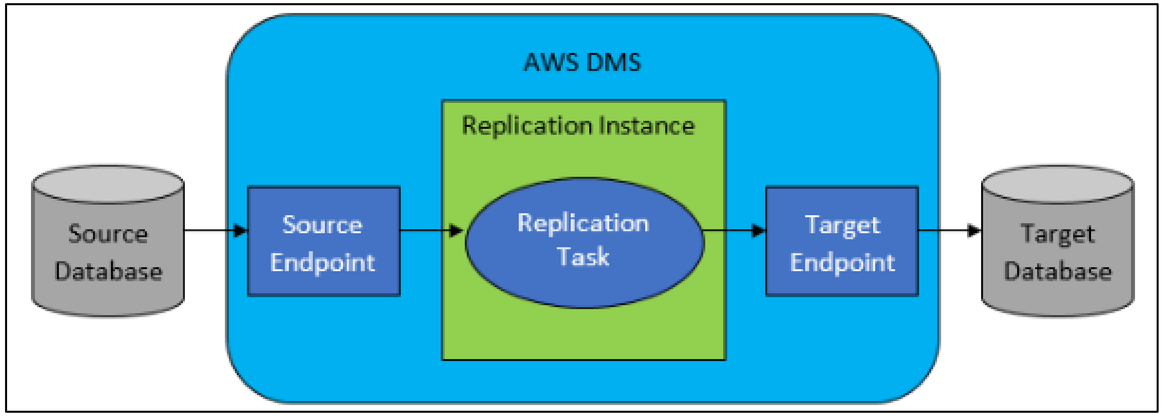
Database Migration Service (DMS)

Migrating MySQL to AWS Aurora from S3



AWS SCT Highlights

Assessment Report	Converts Schema and Code	Extracts and Migrates DW to Amazon Redshift
• Assessment of migration compatibility of source databases with open-source database engines—Amazon RDS for MySQL, Amazon RDS for PostgreSQL, and Amazon Aurora • Recommends best target engine • Provides detailed level of effort to complete migration	• Attempts to convert all schema and code objects to the target engine, including stored procedures and functions • Scans and converts embedded SQL statements in app code • Generates a report with recommendations	• Extracts data through local migration agents • Files are loaded to an Amazon S3 bucket and to Amazon Redshift • Netezza • Vertica • Greenplum • Teradata • Oracle • SQL Server

Description
AWS cloud service that makes it easy to migrate relational databases, data warehouses, NoSQL databases, and other types of data stores
Perform one-time migrations, and you can replicate ongoing changes to keep sources and targets in sync
AWS Schema Conversion Tool (AWS SCT) to translate your database schema to the new platform

https://aws.amazon.com/blogs/database/new-aws-dms-and-aws-snowball-integration-enables-mass-database-migrations-and-migrations-of-large-databases/

AWS RDS Demo

Post Migration Validation and Limitations
Additional article for reference

Homogenous Database Migration

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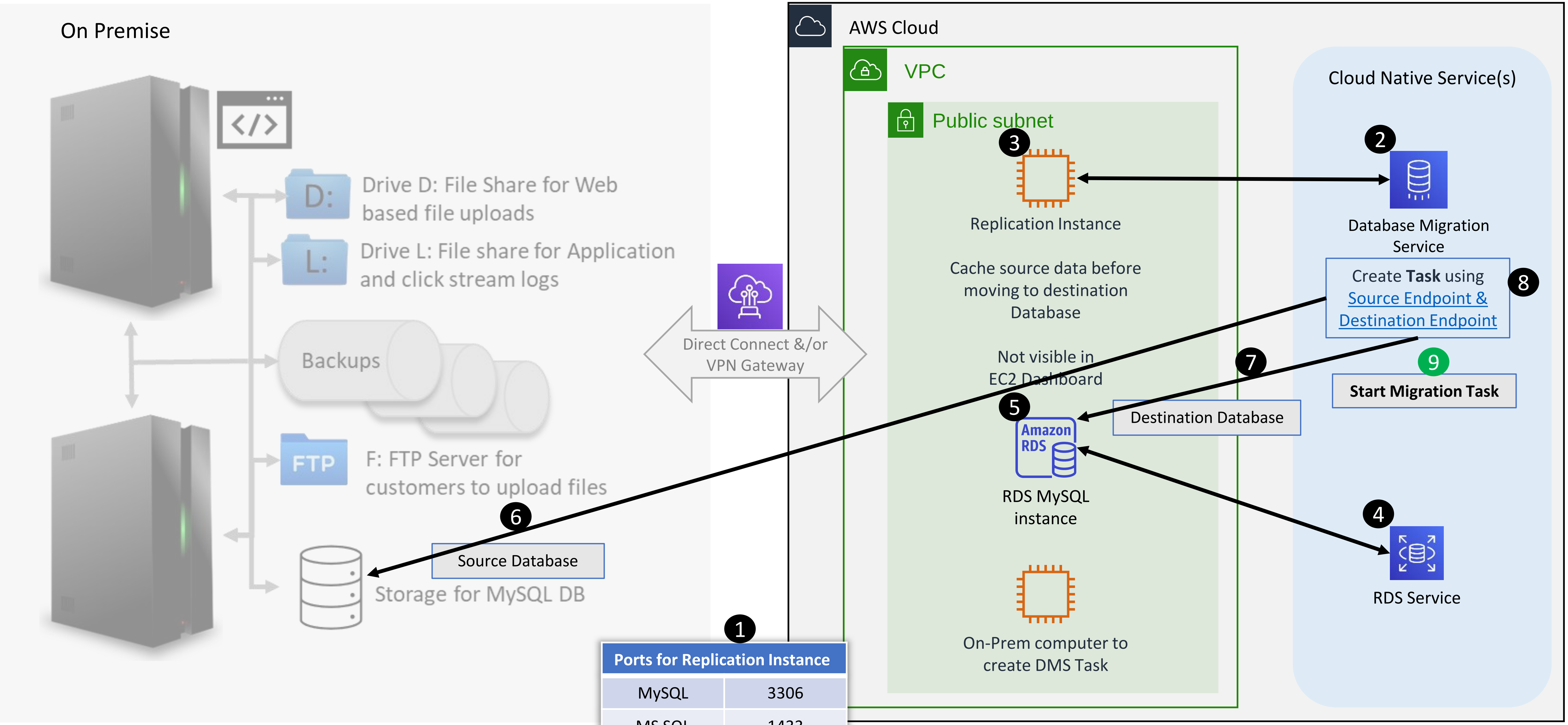

Ports for Replication Instance	
MySQL	3306
MS SQL	1433

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Source.MySQL.html
https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Source.html

Heterogenous Database Migration

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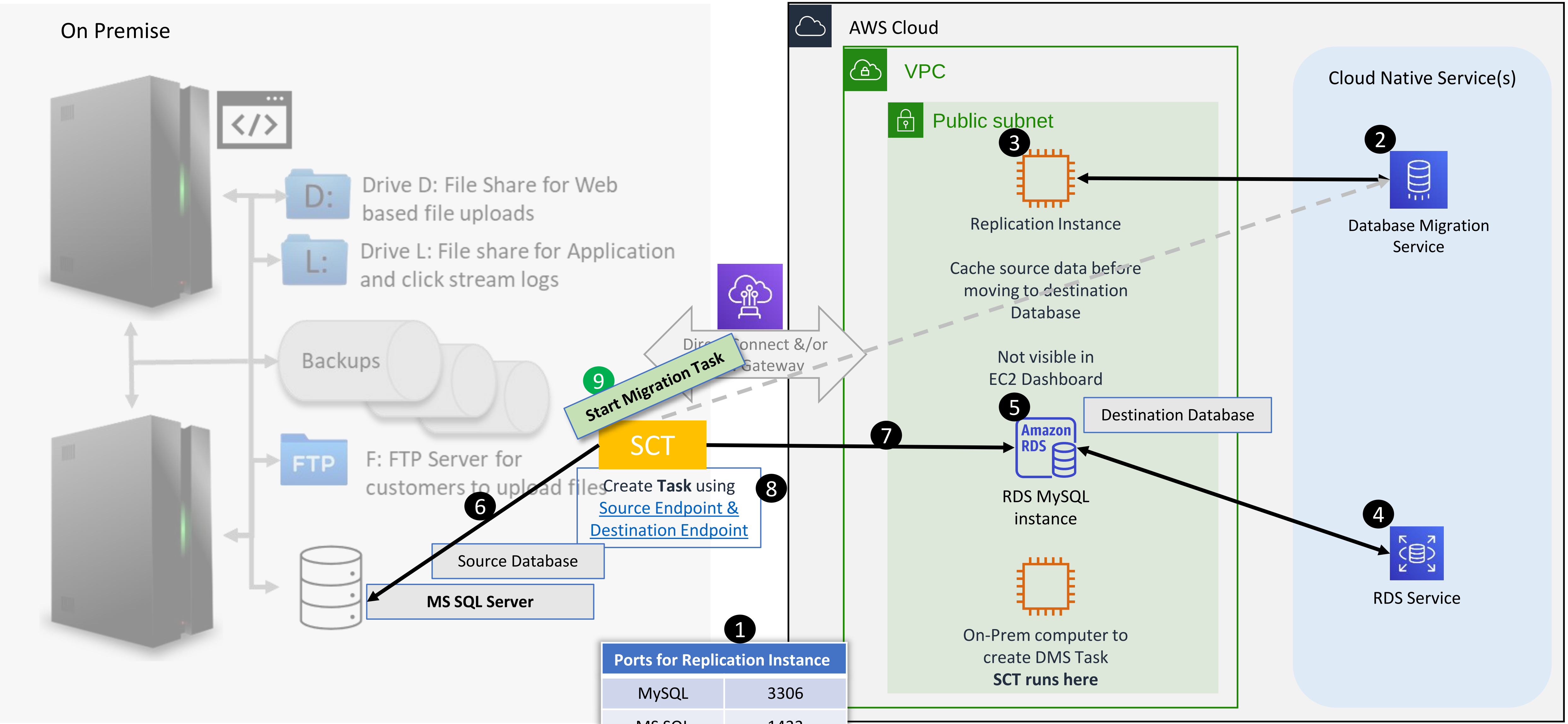

Ports for Replication Instance	
MySQL	3306
MS SQL	1433

Application Migration Process and Tools

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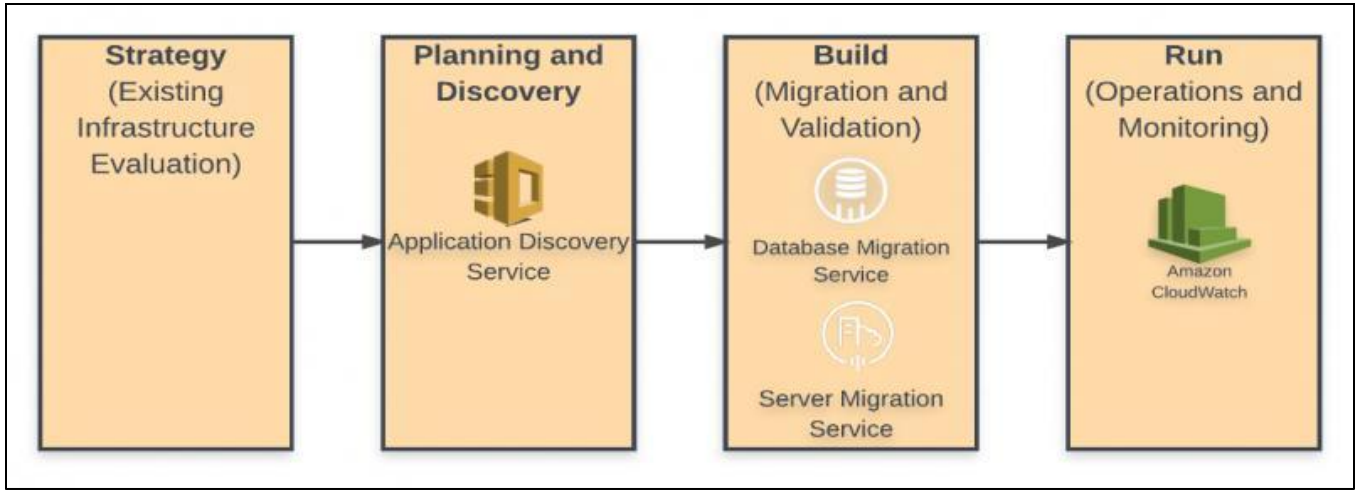
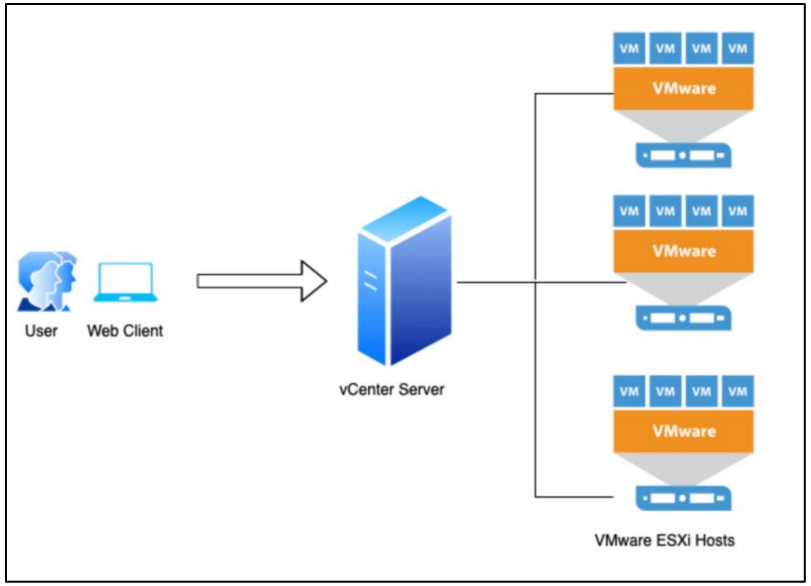
- [Application Discovery Service]
- Application Migration Hub
- Server Migration Service ([SMS])





Description
Helps in planning application migration projects by automatically identifying servers, virtual machines (VMs), software, and software dependencies running in your on-premises data centers
Discovery **Connectors** used VMware vCenter Server. VM Utilization metric collection.
Discovery **Agent** collects richer set of data than the discovery connector, need to install on one or more non-VM hosts in a data center. The agent captures infrastructure and application information, including an inventory of installed software applications, system and process performance, resource utilization, and network dependencies between workloads. See Data Collected by the Discovery Agent.
Import collected data in Migration Hub to associate applications with servers and track migrations. Helps identify or recommend **Instance Type**

Application Migration Process and Tools

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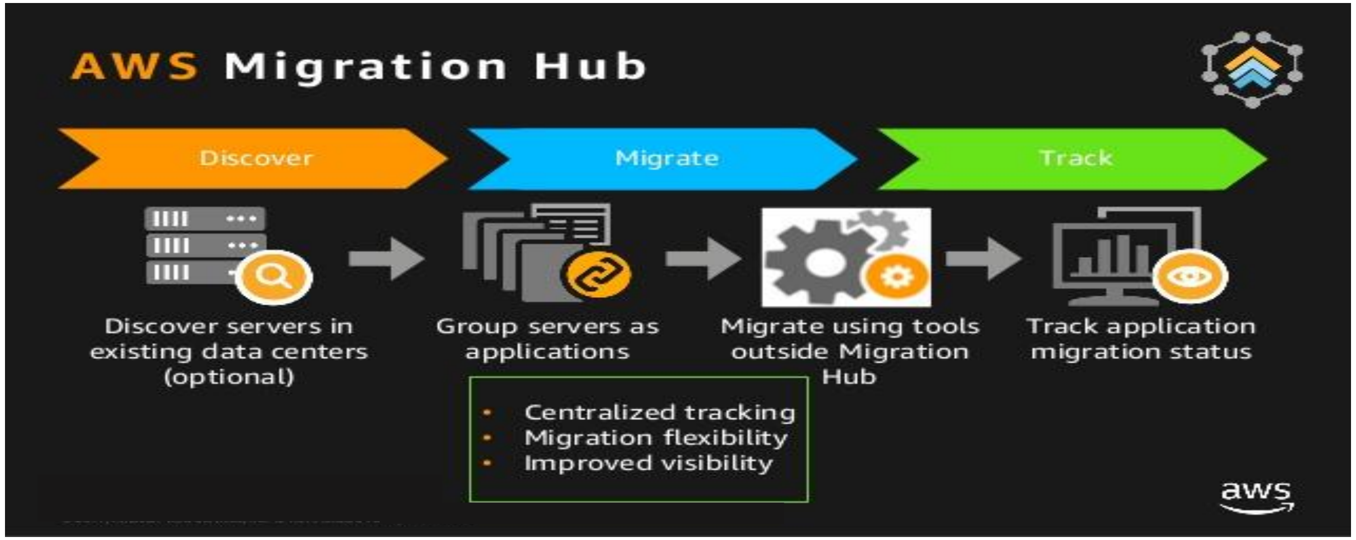
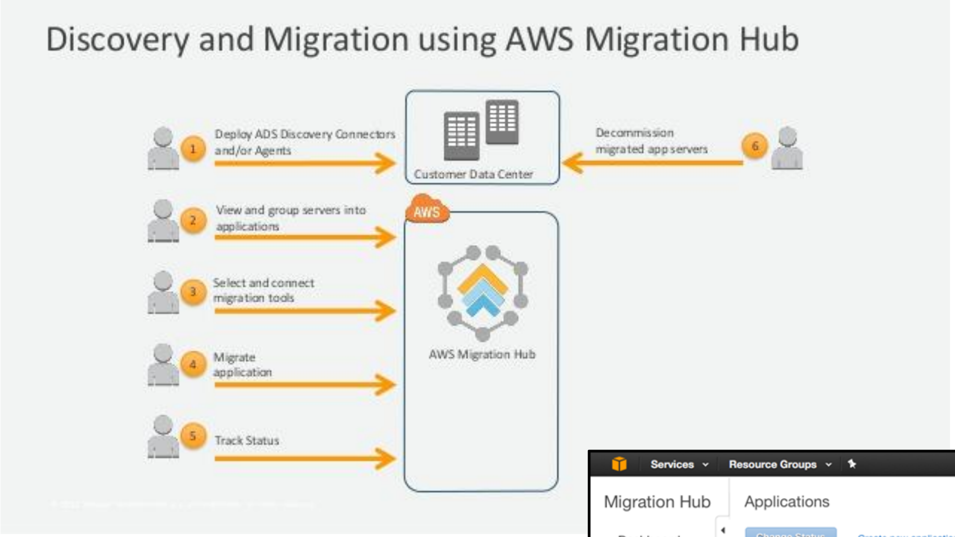
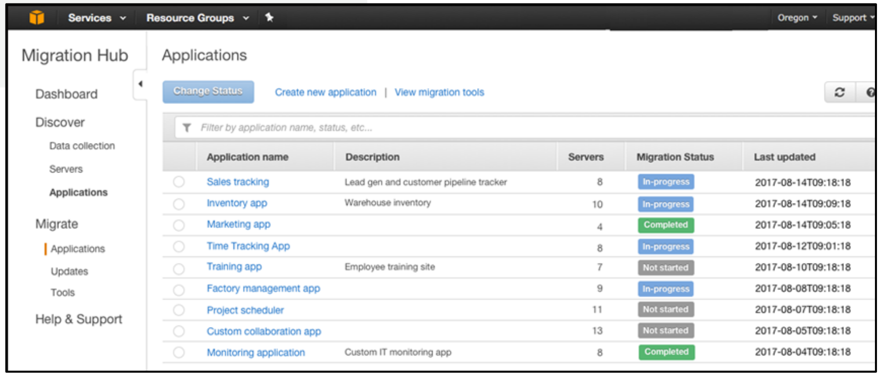
- Application Discovery Service
- Application Migration Hub
- Server Migration Service (SMS)

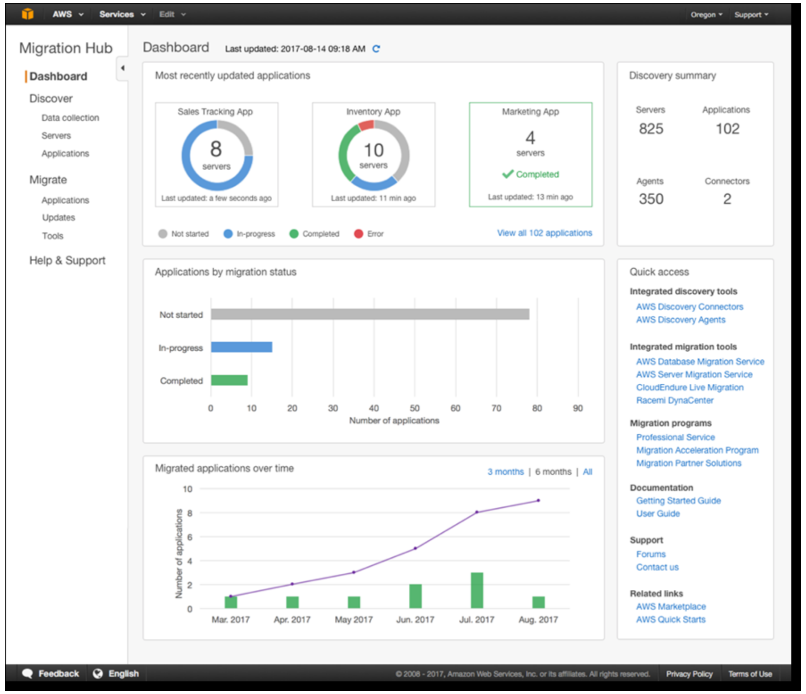




Description
Centralized Tracking
Migration Schedule
Improved Visibility

This is Not a complete answer to migration process.
Complex inter dependent applications needs manual tracking or some how grouping under Application or servers, status updates done manually or via CLI.





Application Migration Process and Tools

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- Application Discovery Service
- Application Migration Hub
- [Server Migration Service (SMS)]

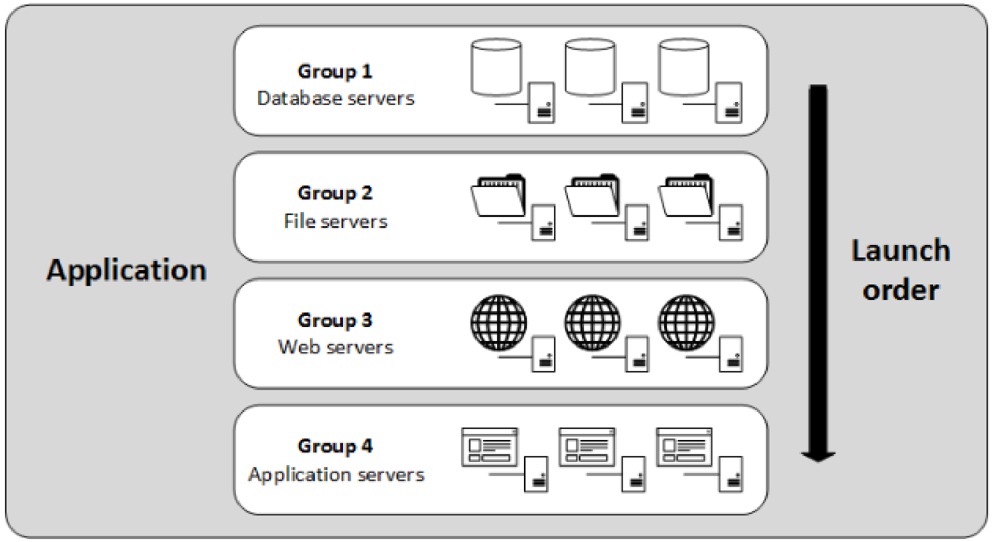
    VMWare VMs migration steps

    Server Migration Pre-reqs

    Evaluate your Migration Needs

Automate via CloudEndure Migration

Automate via RiverMeadow



Description
Automated migration of multi-server application stacks from your on-premises data center to Amazon EC2.
Replicates all the servers in an application as AMIs and generates an AWS CloudFormation template to launch them in a coordinated fashion.
After your servers are organized into applications and launch groups, you can specify a replication frequency, provide configuration scripts, and configure a target VPC in which to launch them.
After 90 days, replication job is automatically terminated. You can request an extension from AWS Support